

EE-102: Electric Circuit Analysis

LECTURE NO 8:

Inductance and Inductors

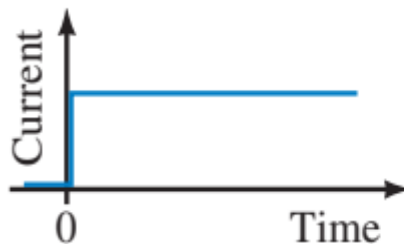
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Organization of Lecture

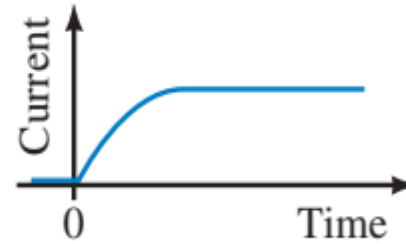
1. Introduction
2. Magnetic Field
3. Inductance
4. Induced Voltage (v_L)
5. R - L Transients: The Storage Phase
6. Initial Conditions
7. R - L Transients: The Release Phase
8. Thevenin's Equivalent: $\tau = L/R_{Th}$
9. Average Induced Voltage: v_{Lav}
10. Inductors in Series and Parallel
11. Steady-State Conditions
12. Energy stored by an Inductor

1. Introduction (1/3)

- In many ways, the *inductor is the dual of the capacitor*:
 - Capacitor stores energy in form of electric field.
 - Inductor stores energy in form of magnetic field.
 - Capacitor opposes change in voltage.
 - Inductor opposes change (sudden change) in current.
- Like the capacitor, the inductor exhibits its true characteristics only when a change in voltage or current is made in the network.



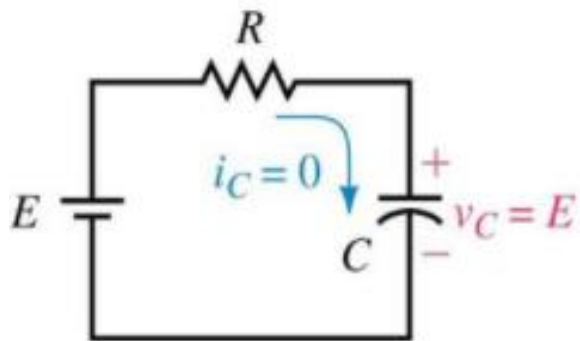
(a) Current cannot jump from one value to another like this



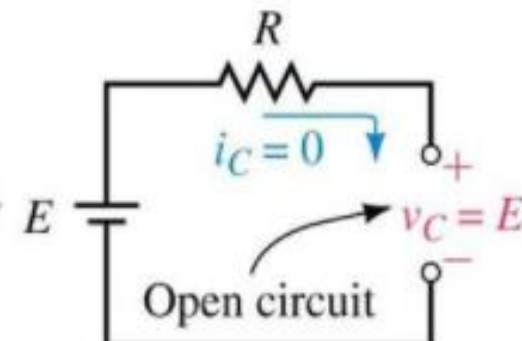
(b) Current must change smoothly with no abrupt jumps

1. Introduction (2/3)

- A capacitor can be replaced by an open-circuit equivalent under steady-state conditions (steady-state DC).
- In steady state DC, the rate of change of voltage is zero.
- A capacitor looks like an open circuit with voltage $v_C = E$ as steady state DC.



(a) $v_C = E$ and $i_C = 0$



(b) Equivalent circuit for the capacitor

Fig. 1.

1. Introduction (3/3)

- You will see in this lecture that *an inductor can be replaced by a short-circuit equivalent under steady-state conditions*. Finally, you will learn that while resistors dissipate the power delivered to them in the form of heat, ideal capacitors store the energy delivered to them in the form of an electric field. Inductors, in the ideal sense, are like capacitors in that they also store the energy delivered to them-but in the form of a magnetic field.

2. Magnetic Field (1/8)

- Magnetism plays an integral part in almost every electrical device used today in industry, research, or the home. Generators, motors, transformers, circuit breakers, televisions, computers, tape recorders, and telephones all employ magnetic effects to perform a variety of important tasks.
- The compass, relies on a *permanent magnet* for indicating direction. A permanent magnet is made of a material, such as steel or iron, that remains magnetized for long periods of time without the need for an external source of energy.
- In 1820, the Danish physicist Hans Christian Oersted discovered that the needle of a compass deflects if brought near a current-carrying conductor. This was the first demonstration that electricity and magnetism were related.

2. Magnetic Field (2/8)

- In subsequent years, others, such as Michael Faraday, Karl Friedrich Gauss, and James Clerk Maxwell, continued to experiment in this area and developed many of the basic concepts of *electromagnetism*- magnetic effects induced by the flow of charge, or current.
- A magnetic field exists in the region surrounding a permanent magnet, which can be represented by **magnetic flux lines** as shown in Fig. 2.

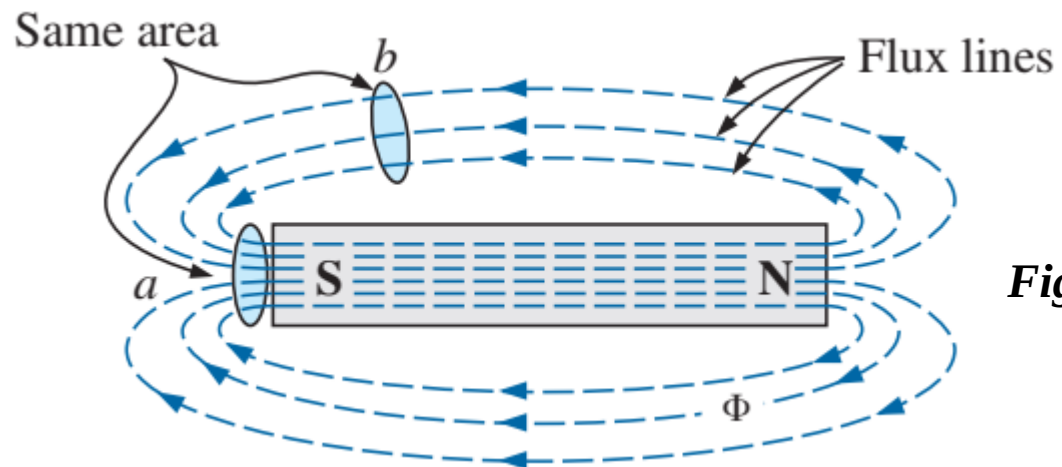


Fig. 2.

2. Magnetic Field (3/8)

- The magnetic flux lines radiate from the north pole to the south pole, returning to the north pole through the metallic bar as shown in Fig. 2.
- The strength of a magnetic field in a particular region is directly related to the density of flux lines in that region. In Fig. 2, for example, the magnetic field strength(magnetic flux lines) at point a is twice that at point b since twice as many magnetic flux lines are associated with the perpendicular plane at point a than at point b .
- If unlike poles of two permanent magnets are brought together, the magnets attract, and the flux distribution is as shown in Fig. 3, next slide . If like poles are brought together, the magnets repel, and the flux distribution is as shown in Fig. 4, next slide.

2. Magnetic Field (4/8)

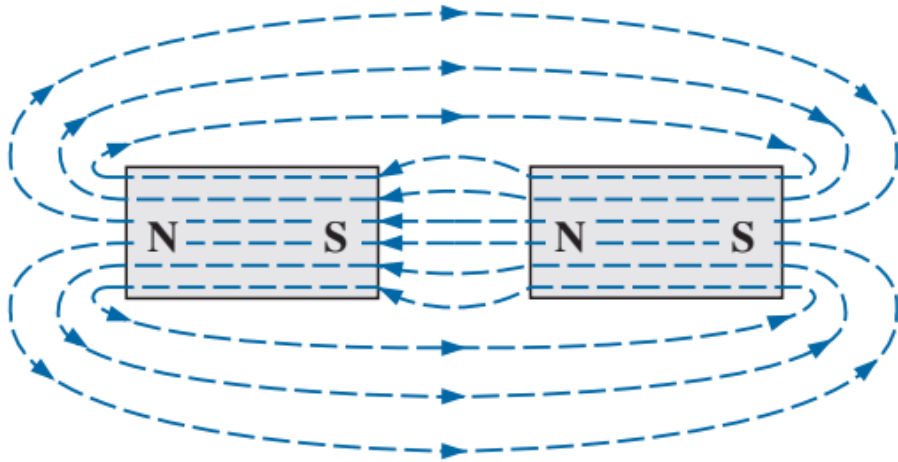


Fig. 3.

Flux distribution for two adjacent, opposite poles.

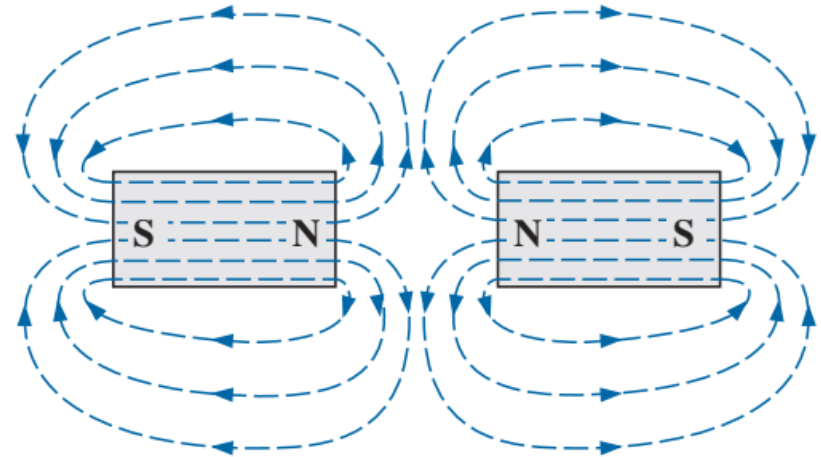


Fig. 4.

Flux distribution for two adjacent, like poles.

2. Magnetic Field (5/8)

- If a nonmagnetic material, such as glass or copper, is placed in the flux paths surrounding a permanent magnet, an almost unnoticeable change occurs in the flux distribution (Fig. 5). However, if a magnetic material, such as soft iron, is placed in the flux path, the flux lines pass through the soft iron rather than the surrounding air because flux lines pass with greater ease through magnetic materials than through air.

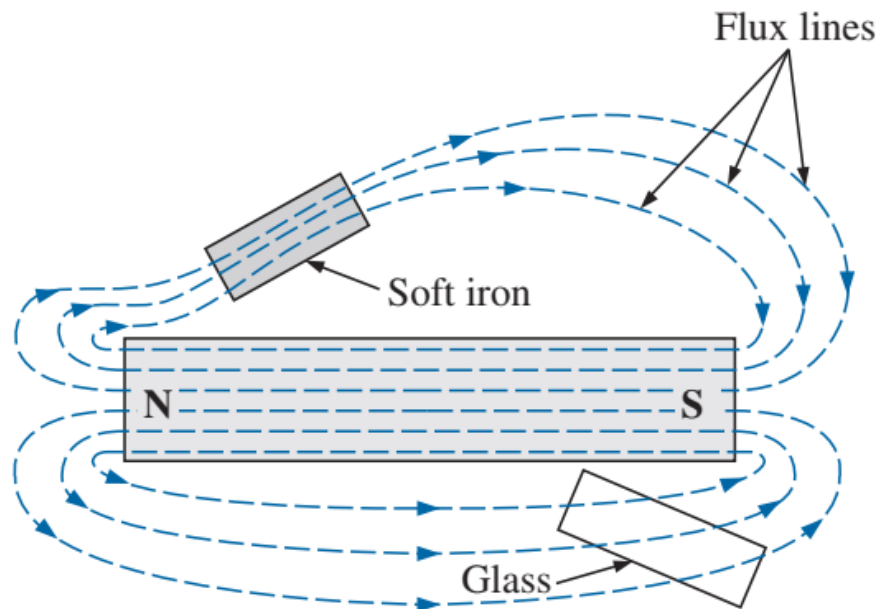


Fig. 5.

2. Magnetic Field (6/8)

- In the SI system of units, magnetic flux is measured in *webers (Wb)*. The applied symbol is the capital Greek letter phi, Φ . The number of flux lines per unit area, called the flux density, is denoted by the capital letter B and is measured in *teslas (T)*. In equation form:

$$B = \frac{\Phi}{A}$$

$$B = \text{Wb/m}^2 = \text{teslas (T)}$$

$$\Phi = \text{webers (Wb)} \quad \dots\dots(1)$$

$$A = \text{m}^2$$

- Where Φ is the number of flux lines passing through area A in Fig. 6.

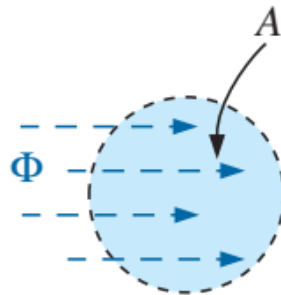


Fig. 6.

2. Magnetic Field (7/8)

- The flux density, **B** of an electromagnet is directly related to the number of turns of, and current through, the coil. The product of the two, called the **magnetomotive force**, is measured in *ampere-turns* (At) as defined by:

$$\mathcal{F} = NI \quad (\text{ampere-turns, At}) \quad \dots\dots(2)$$

- Another factor that affects the magnetic field strength is the type of core used. Materials in which magnetic flux lines can readily be set up are said to be magnetic and to have a high **permeability**. The permeability (represented by the Greek letter mu, **μ**) of a material is a measure of the ease with which magnetic flux lines can be established in the material. There is a specific number associated with the permeability of air:

$$\mu_o = 4\pi \times 10^{-7} \text{ Wb/A} \cdot \text{m} \quad \dots\dots(3)$$

2. Magnetic Field (8/8)

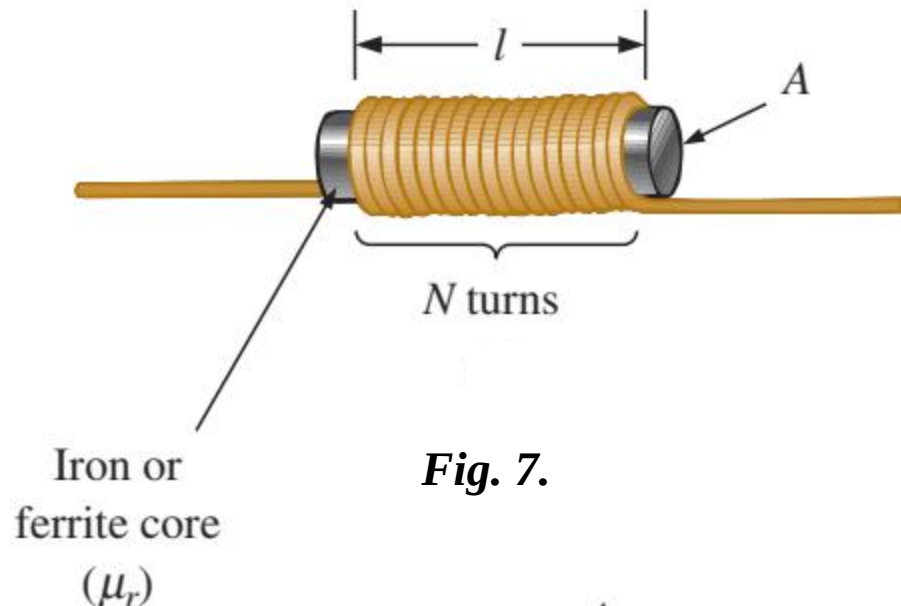
- Practically speaking, the permeability of all nonmagnetic materials, such as copper, aluminium, wood, glass, and air, is the same as that for free space.
- Magnetic materials, such as iron, nickel, steel, cobalt, and alloys of these metals, have permeabilities hundreds and even thousands of times that of free space. Materials with very high permeabilities are referred to as **ferromagnetic**.
- The ratio of the permeability of a material to that of free space is called its **relative permeability**; that is,

$$\mu_r = \frac{\mu}{\mu_o} \dots\dots(4)$$

- In general, for ferromagnetic materials, $\mu_r \geq 100$, and for nonmagnetic materials, $\mu_r = 1$.

3. Inductance (1/6)

- An inductor is a passive element (*as it stores energy in a magnetic field*) designed to store energy in its magnetic field.
 - Inductors consist of a coil of wire, often wound around a core of high magnetic permeability.
- *1 henry is the inductance level that will establish a voltage of 1 volt across the coil due to a change in current of 1 A/s through the coil.*



3. Inductance (2/6)

Inductor Construction

- The level of inductance has construction sensitivities in that it is dependent on the area (A) within the coil, the length of the unit (l), and the permeability of the core material (μ_r). It is also sensitive to the number of turns of wire in the coil (N) as dictated by the following equation and defined in Fig. 8
- Inductance (L) is measured in Henries (H).

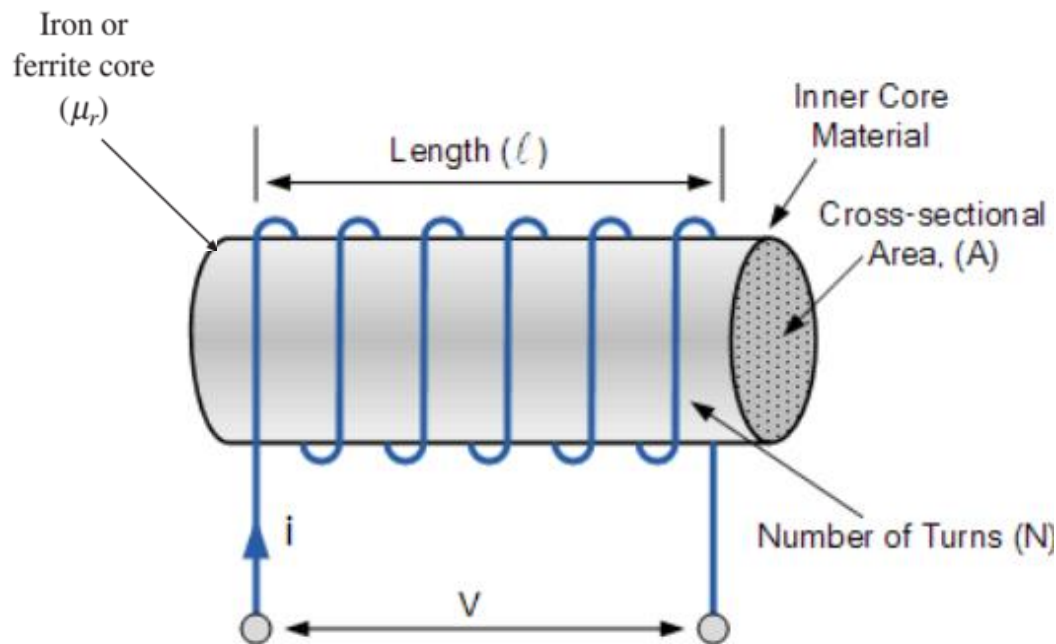


Fig. 8.

3. Inductance (3/6)

Inductor Construction (continued...)

$$L = \frac{\mu N^2 A}{l}$$

μ = permeability (Wb/A · m)

N = number of turns (t)

A = m²

l = m

L = henries (H)

.....(5)

- Substituting $\mu = \mu_r \mu_o$ for the permeability results in equation 5 we get,

$$L = \frac{\mu_r \mu_o N^2 A}{l} \quad \text{.....(6)}$$

$$L = 4\pi \times 10^{-7} \frac{\mu_r N^2 A}{l}$$

(henries, H)(7)

3. Inductance (4/6)

Inductor Construction (continued...)

- If we break out the relative permeability as (Refer to equation 6)

$$L = \mu_r \left(\frac{\mu_o N^2 A}{l} \right)$$

we obtain the following useful equation:

$$\boxed{L = \mu_r L_o} \quad \dots\dots(8)$$

The inductance of an inductor with a ferromagnetic core is μ_r times the inductance obtained with an air core.

3. Inductance (5/6)

EXAMPLE 1 For the air-core coil in Fig. 9:

- Find the inductance.
- Find the inductance if a metallic core with $\mu_r = 2000$ is inserted in the coil.

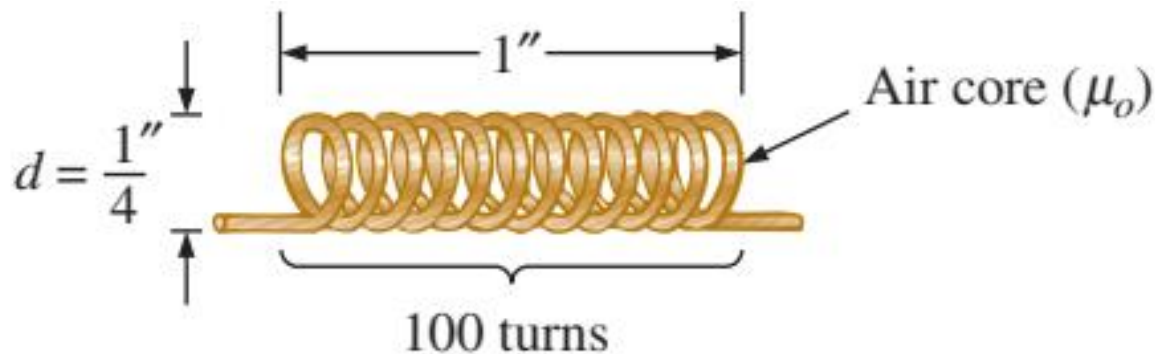


Fig. 9.

3. Inductance (6/6)

Solutions:

$$\text{a. } d = \frac{1}{4} \cancel{\text{in.}} \left(\frac{1 \text{ m}}{39.37 \cancel{\text{in.}}} \right) = 6.35 \text{ mm}$$

$$A = \frac{\pi d^2}{4} = \frac{\pi (6.35 \text{ mm})^2}{4} = 31.7 \mu\text{m}^2$$

$$l = 1 \cancel{\text{in.}} \left(\frac{1 \text{ m}}{39.37 \cancel{\text{in.}}} \right) = 25.4 \text{ mm}$$

$$L = 4\pi \times 10^{-7} \frac{\mu_r N^2 A}{l} \quad \dots \text{Using eq. (7)}$$

$$= 4\pi \times 10^{-7} \frac{(1)(100 \text{ t})^2 (31.7 \mu\text{m}^2)}{25.4 \text{ mm}} = \mathbf{15.68 \mu\text{H}}$$

$$\text{b. Equation 8: } L = \mu_r L_o = (2000)(15.68 \mu\text{H}) = \mathbf{31.36 \text{ mH}}$$

Part (a):

For non-magnetic material: $\mu_r = 1$

4. Induced Voltage v_L (1/5)

- Before analyzing the response of inductive elements to an applied dc voltage, we must introduce a number of laws and equations that affect the transient response.
- The first, referred to as *Faraday's law of electromagnetic induction*, is one of the most important in this field.
- If we move a conductor through a magnetic field so that it cuts magnetic lines of flux as shown in Fig.10 next slide, a voltage is induced across the conductor that can be measured with a sensitive voltmeter. The faster you move the conductor through the magnetic flux, the greater the induced voltage.
- Note that the direction in which you move the conductor through the field determines the polarity of the induced voltage. Also, if you move the conductor through the field at right angles to the magnetic flux, you generate the maximum induced voltage.

4. Induced Voltage v_L (2/5)

- Moving the conductor parallel with the magnetic flux lines results in an induced voltage of zero volts since magnetic lines of flux are not crossed.
- If we go a step further and move a coil of N turns through the magnetic field as shown in Fig. 11 next slide, a voltage will be induced across the coil as determined by *Faraday's law*:

$$e = N \frac{d\phi}{dt} \quad (\text{volts, V}) \dots\dots(9)$$

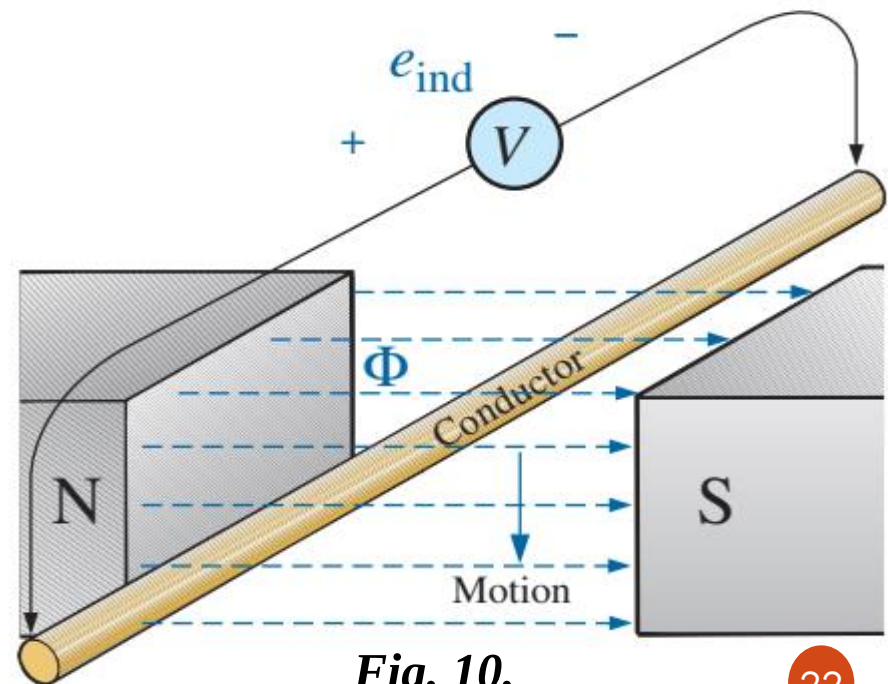


Fig. 10.

4. Induced Voltage v_L (3/5)

- It is very important to note in Fig. 12 that the polarity of the induced voltage across the coil is such that it opposes the increasing level of current in the coil.
- The quicker the change in current through the coil, the greater is the opposing induced voltage to squelch the attempt of the current to increase in magnitude.

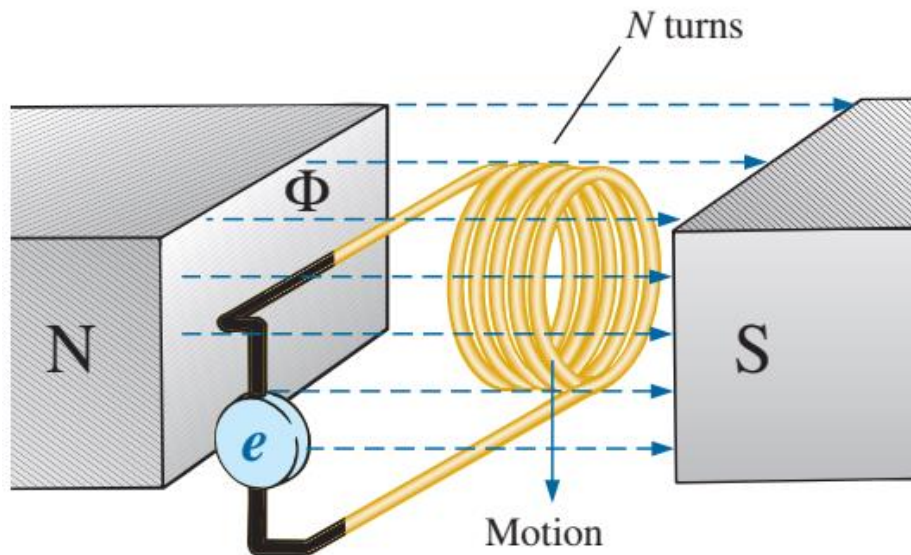


Fig. 11.

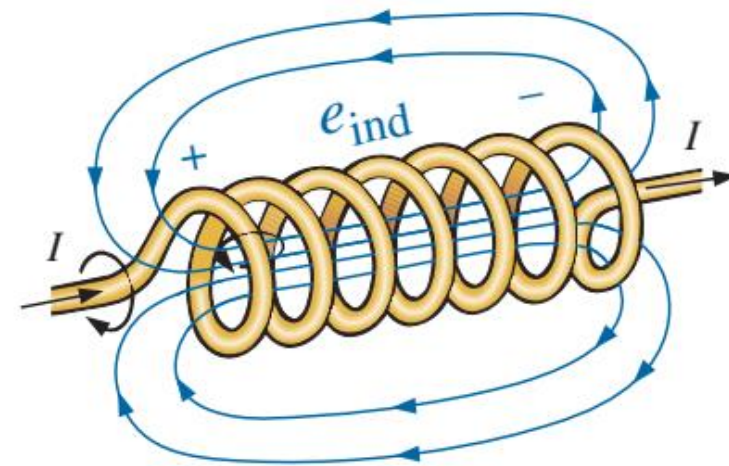


Fig. 12.

4. Induced Voltage v_L (4/5)

- The effect in Fig. 12 is a result of an important law referred to as **Lenz's law**, which states that- *an induced effect is always such as to oppose the cause that produced it.*
- The inductance of a coil can also be determined as :

$$L = N \frac{d\phi}{di_L} \quad (\text{henries, H}) \quad \dots\dots(10)$$

- Similarly the induced voltage can also be determined as (as per lenz's law):

$$e_L = L \frac{di_L}{dt} \quad (\text{volts, V}) \quad \dots\dots(11)$$

4. Induced Voltage v_L (5/5)

- The following notation is used for the induced voltage across an inductor:

$$\boxed{v_L = L \frac{di_L}{dt}} \quad (\text{volts, V}) \quad \text{.....(12)}$$

- Equation 12, clearly states that, the larger the inductance and/or the more rapid the change in current through a coil, the larger will be the induced voltage across the coil.
- If the current through the coil fails to change with time, the induced voltage across the coil will be zero. When the transient phase has passed, $di_L/dt = 0$, and the induced voltage across the coil is:

$$v_L = L \frac{di_L}{dt} = L(0) = 0 \text{ V}$$

5. R-L Transients: The Storage Phase (1/13)

- A great number of similarities exist between the analyses of inductive and capacitive networks. That is, *what is true for the voltage of a capacitor is also true for the current of an inductor, and what is true for the current of a capacitor can be matched in many ways by the voltage of an inductor.* The storage waveforms have the same shape, and time constants are defined for each configuration.
- The circuit in Fig. 13 next slide is used to describe the storage phase. Note that it is the same circuit used to describe the charging phase of capacitors, with a simple replacement of the capacitor by an ideal inductor.
- Throughout the analysis, it is important to remember that energy is stored in the form of an electric field between plates of a capacitor. For inductors, on the other hand, energy is stored in the form of a magnetic field.

5. R - L Transients: The Storage Phase (2/13)

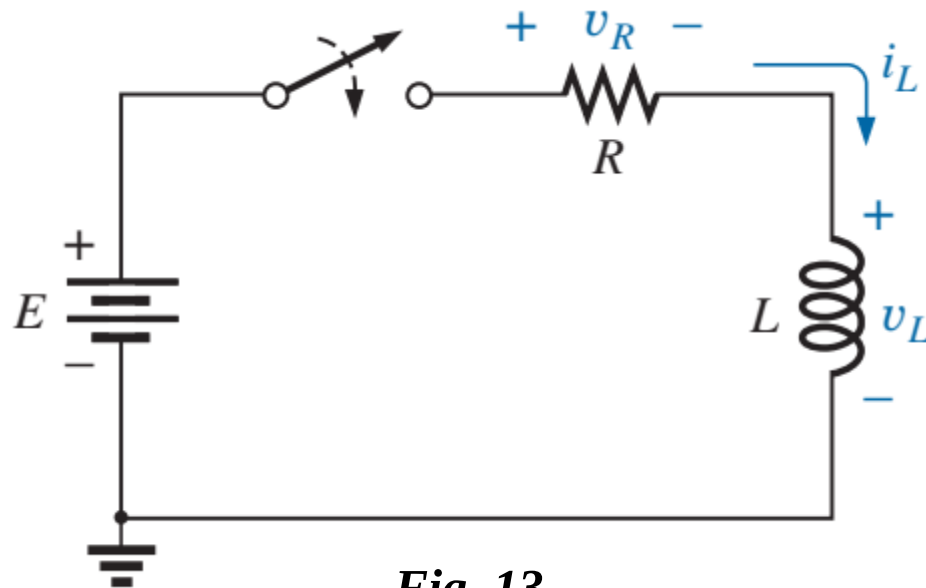
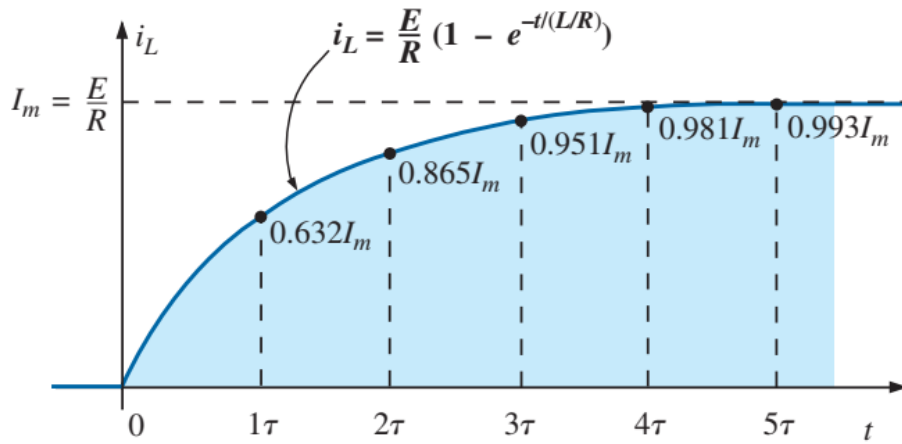


Fig. 13.

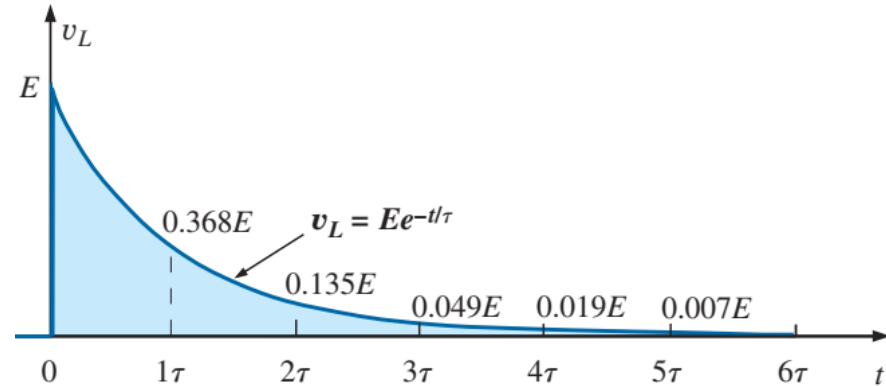
5. R - L Transients: The Storage Phase (3/13)

- At the instant the switch is closed, the choking action of the coil prevents an instantaneous change in current through the coil, resulting in:
 - $i_L = 0A$. (as shown in Fig. 14(a), next slide)
- The absence of a current through the coil and circuit at the instant the switch is closed results in zero volts across the resistor as determined by:
 - $v_R = i_R R = I_L R = (0A)R = 0V$. As $i_R = i_L$ in series circuit(as shown in Fig, 14(c), next slide).
- Applying Kirchhoff's voltage law on Fig. 13 around the closed loop results in E volts across the coil at the instant the switch is closed,
 - $E - v_R - v_L = 0$
 - $E - v_R = v_L$
 - $E - 0V = v_L$ (As $v_R = 0V$ at the time when switch is closed).
 - $v_L = E$ (as shown in Fig, 14(b), next slide)

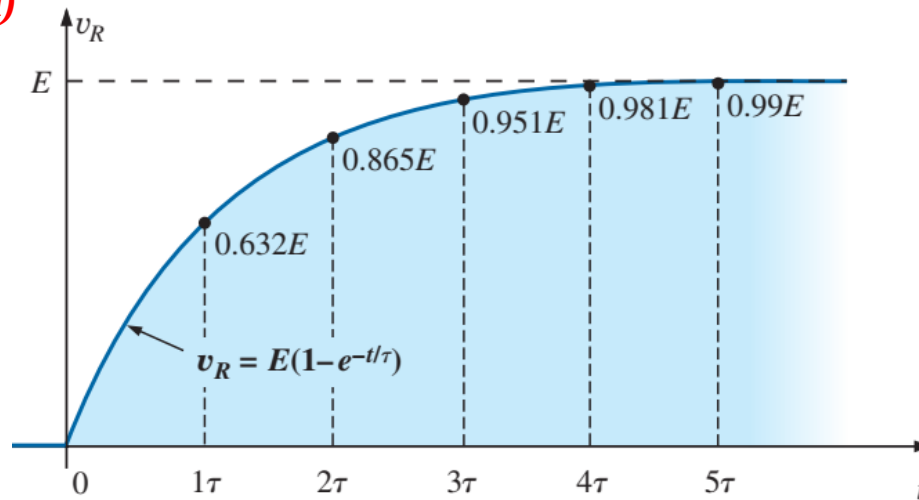
5. R - L Transients: The Storage Phase (4/13)



(a)



(b)



(c)

Fig. 14.

5. *R-L* Transients: The Storage Phase (5/13)

- Initially, the current increases very rapidly, as shown in Fig. 14(a) and then at a much slower rate as it approaches its steady-state value determined by the parameters of the network (E/R). The voltage across the resistor rises at the same rate because: $v_R = i_R R = i_L R$.
- Finally when the current reaches its steady-state value of E/R amperes, the current through the coil ceases to change, and the voltage across the coil drops to zero volts.
- At any instant of time, the voltage across the coil can be determined using Kirchhoff's voltage law in the following manner: $v_L = E - v_R$.

5. *R-L* Transients: The Storage Phase (6/13)

- The equation for the transient response of the current through an inductor is:

$$i_L = \frac{E}{R}(1 - e^{-t/\tau}) \quad (\text{amperes, A}) \dots\dots(13)$$

with the time constant now defined by

$$\tau = \frac{L}{R} \quad (\text{seconds, s}) \dots\dots(14)$$

- $i_L = I_m(1 - e^{-t/\tau})$
- $I_m = E/R$
- $\tau = L/R$

5. *R-L* Transients: The Storage Phase (7/13)

- If we keep R constant and increase L , the ratio L/R increases, and the rise time of 5τ increases as shown in Fig. 15 next slide for increasing levels of L . The change in transient response is expected because the higher the inductance level, the greater is the choking action on the changing current level, and the longer it will take to reach steady-state conditions.
- The equation for the voltage across the coil is:

$$v_L = Ee^{-t/\tau} \quad (\text{volts, V}) \dots\dots(15)$$

and the equation for the voltage across the resistor is

$$v_R = E(1 - e^{-t/\tau}) \quad (\text{volts, V}) \dots\dots(16)$$

5. R - L Transients: The Storage Phase (8/13)

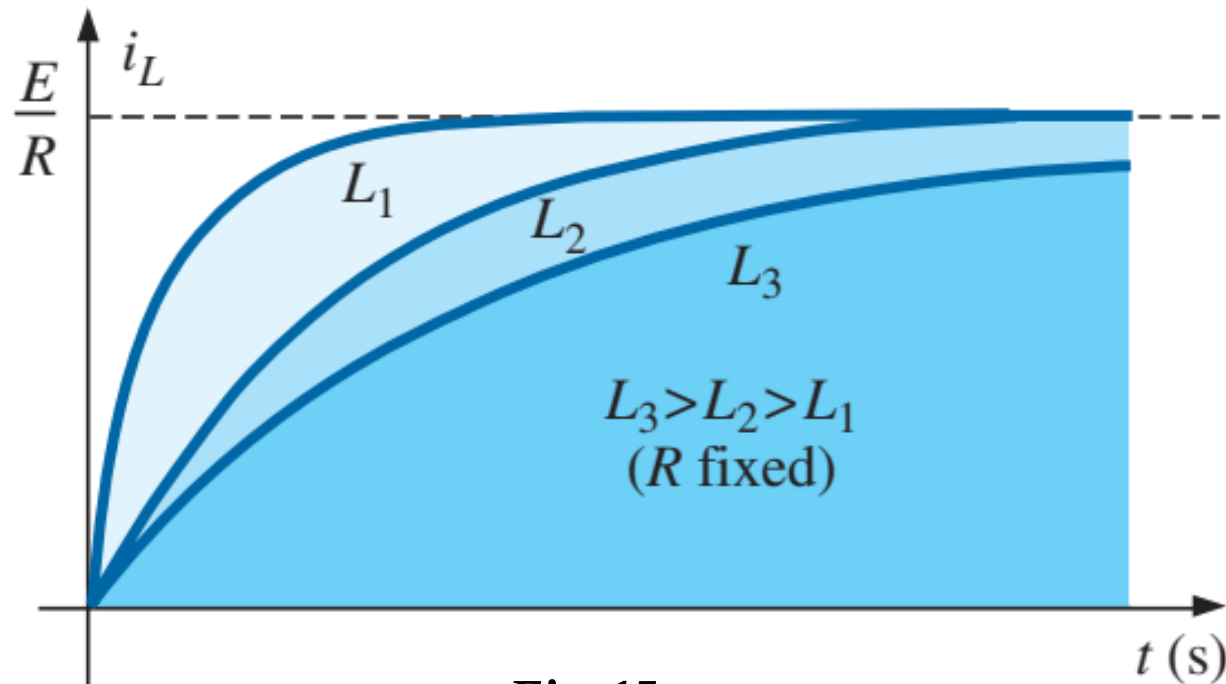
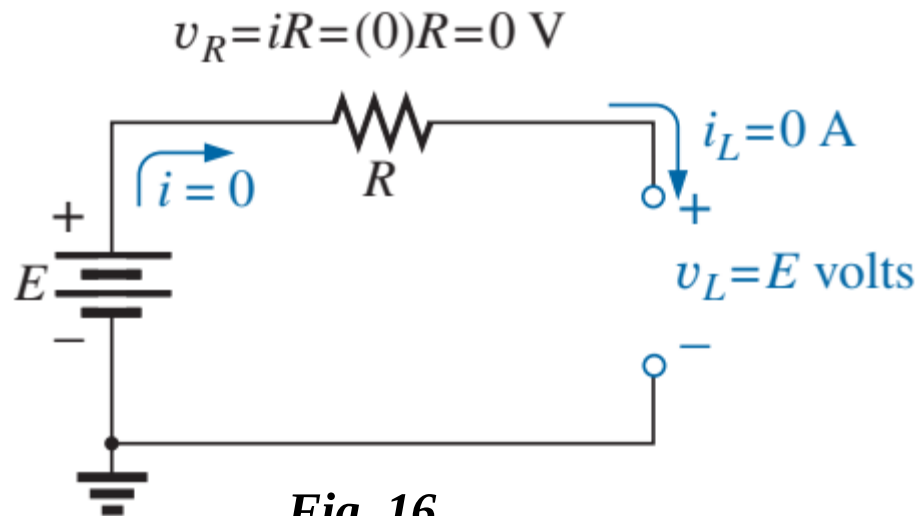


Fig. 15.

5. R - L Transients: The Storage Phase (9/13)

- The current cannot change instantaneously in an inductive network.
- If we examine the conditions that exist at the instant the switch is closed, we find that the voltage across the coil is E volts, although the current is zero amperes as shown in Fig. 16. In essence, therefore:
- *The inductor takes on the characteristics of an open circuit at the instant the switch is closed.*



5. R-L Transients: The Storage Phase (10/13)

- The inductor takes on the characteristics of a short circuit when steady-state conditions have been established.*

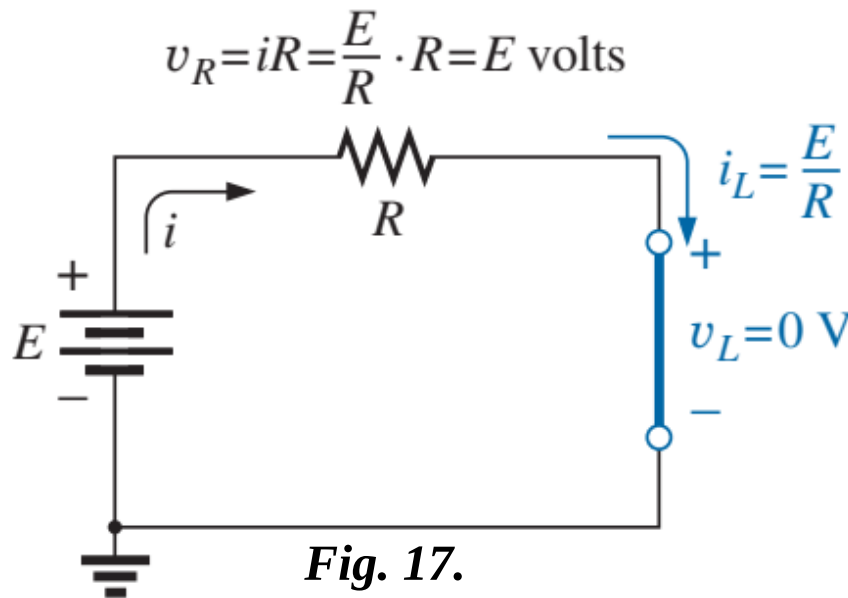
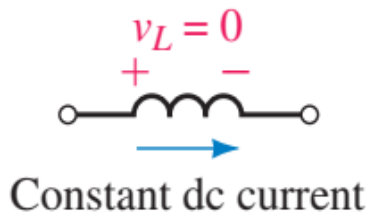
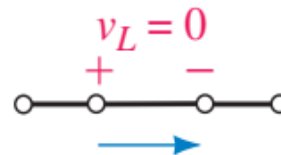


Fig. 17.



(a) Since the field is constant, induced voltage is zero



(b) Equivalent of inductor to dc is a short circuit

5. R - L Transients: The Storage Phase (11/13)

EXAMPLE 2 Find the mathematical expressions for the transient behavior of i_L and v_L for the circuit in Fig. 18 if the switch is closed at $t = 0$ s. Sketch the resulting curves.

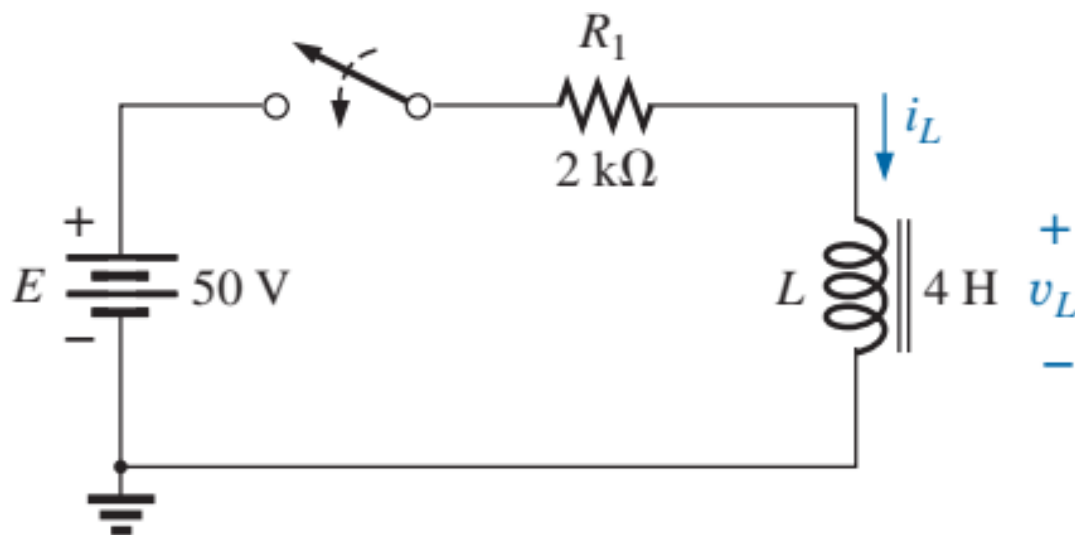


Fig. 18.

5. R - L Transients: The Storage Phase (12/13)

Solution: First, the time constant is determined:

$$\tau = \frac{L}{R_1} = \frac{4 \text{ H}}{2 \text{ k}\Omega} = 2 \text{ ms}$$

Then the maximum or steady-state current is

$$I_m = \frac{E}{R_1} = \frac{50 \text{ V}}{2 \text{ k}\Omega} = 25 \times 10^{-3} \text{ A} = 25 \text{ mA}$$

Substituting into Eq. (13):

$$i_L = 25 \text{ mA} (1 - e^{-t/2\text{ms}})$$

Using Eq. (15):

$$v_L = 50 \text{ V} e^{-t/2\text{ms}}$$

5. R - L Transients: The Storage Phase (13/13)

The resulting waveforms appear in Fig. 19.

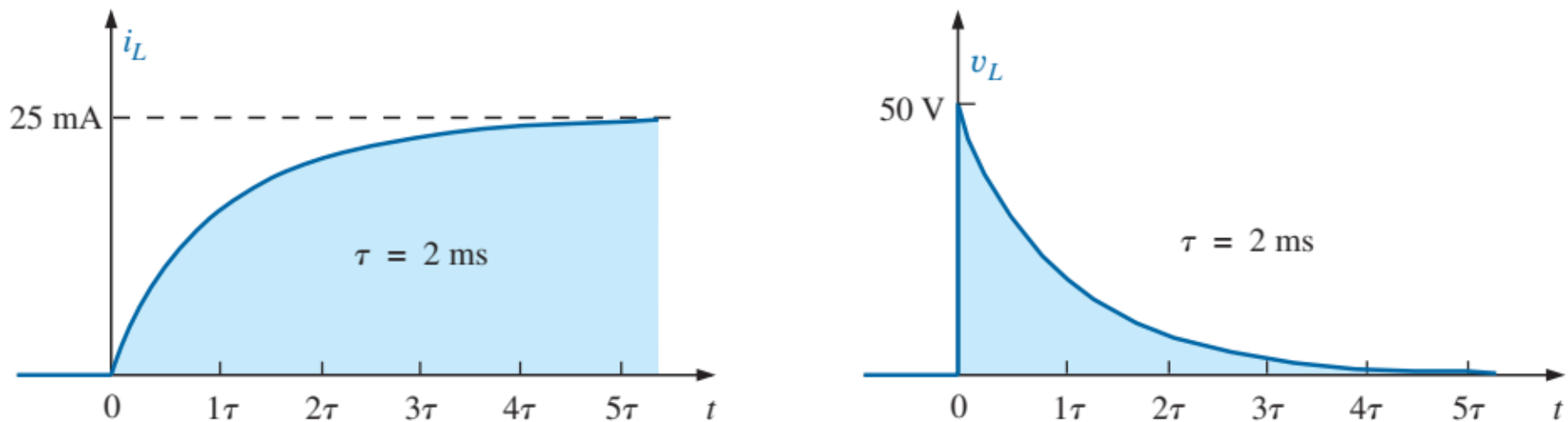


FIG. 19

i_L and v_L for the network in Fig. 18

6. Initial Conditions (1/7)

- Since the current through a coil cannot change instantaneously, the current through a coil begins the transient phase at the initial value established by the network (note Fig. 20) before the switch was closed. It then passes through the transient phase until it reaches the steady-state (or final) level after about five time constants.

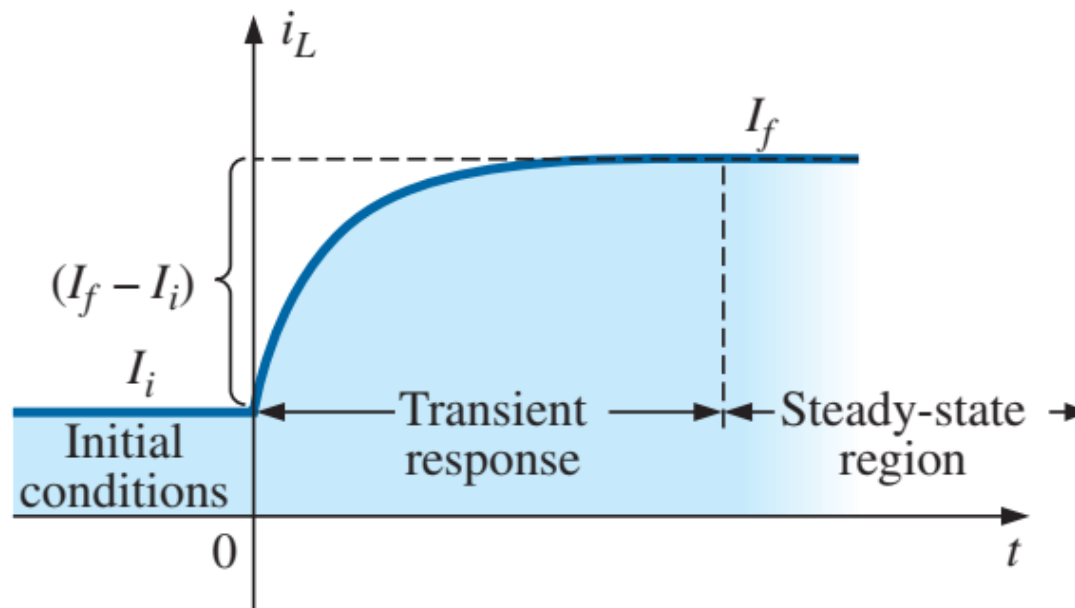


FIG. 20

Defining the three phases of a transient waveform.

6. Initial Conditions (2/7)

Using the transient equation developed in the previous section, we can write an equation for the current i_L for the entire time interval in Fig. 20; that is,

$$i_L = I_i + (I_f - I_i)(1 - e^{-t/\tau})$$

with $(I_f - I_i)$ representing the total change during the transient phase. However, by multiplying through and rearranging terms as

$$\begin{aligned} i_L &= I_i + I_f - I_f e^{-t/\tau} - I_i + I_i e^{-t/\tau} \\ &= I_f - I_f e^{-t/\tau} + I_i e^{-t/\tau} \end{aligned}$$

we find

$$i_L = I_f + (I_i - I_f)e^{-t/\tau} \quad \dots\dots(17)$$

- i_L = Inductor current.
- I_i = Initial Current.
- I_f = Final or Steady-State Current.

6. Initial Conditions (3/7)

EXAMPLE 3 The inductor in Fig. 21 has an initial current level of 4 mA in the direction shown.

- Find the mathematical expression for the current through the coil once the switch is closed.
- Find the mathematical expression for the voltage across the coil during the same transient period.
- Sketch the waveform for each from initial value to final value.

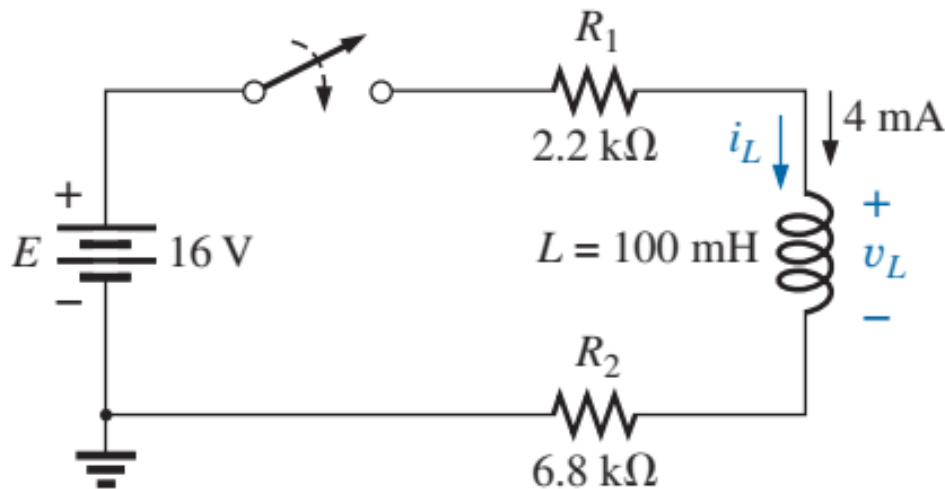


FIG. 21

6. Initial Conditions (4/7)

Solutions:

- a. Substituting the short-circuit equivalent for the inductor results in a final or steady-state current determined by Ohm's law:

$$I_f = \frac{E}{R_1 + R_2} = \frac{16 \text{ V}}{2.2 \text{ k}\Omega + 6.8 \text{ k}\Omega} = \frac{16 \text{ V}}{9 \text{ k}\Omega} = 1.78 \text{ mA}$$

The time constant is determined by

$$\tau = \frac{L}{R_T} = \frac{100 \text{ mH}}{2.2 \text{ k}\Omega + 6.8 \text{ k}\Omega} = \frac{100 \text{ mH}}{9 \text{ k}\Omega} = 11.11 \mu\text{s}$$

Applying Eq. (17):

$$\begin{aligned} i_L &= I_f + (I_i - I_f) e^{-t/\tau} = 1.78 \text{ mA} + (4 \text{ mA} - 1.78 \text{ mA})e^{-t/11.11\mu\text{s}} \\ &= \mathbf{1.78 \text{ mA} + 2.22 \text{ mA}e^{-t/11.11\mu\text{s}}} \end{aligned}$$

6. Initial Conditions (5/7)

- b. Since the current through the inductor is constant at 4 mA prior to the closing of the switch, the voltage (whose level is sensitive only to changes in current through the coil) must have an initial value of 0 V. At the instant the switch is closed, the current through the coil cannot change instantaneously, so the current through the resistive elements is 4 mA. The resulting peak voltage at $t = 0$ s can then be found using Kirchhoff's voltage law as follows:

$$\begin{aligned} V_m &= E - V_{R_1} - V_{R_2} = 16 \text{ V} - (4 \text{ mA})(2.2 \text{ k}\Omega) - (4 \text{ mA})(6.8 \text{ k}\Omega) \\ &= 16 \text{ V} - 8.8 \text{ V} - 27.2 \text{ V} = 16 \text{ V} - 36 \text{ V} = -20 \text{ V} \end{aligned}$$

- $E - V_{R1} - V_m - V_{R2} = 0$ (Apply KVL, clockwise in Fig. 21)
where, V_m : The peak voltage across inductor at time $t = 0$ seconds.
- $V_m = E - V_{R1} - V_{R2}$
- $V_{R1} = I_i R_1$
- $V_{R2} = I_i R_2$

6. Initial Conditions (6/7)

Note the minus sign to indicate that the polarity of the voltage v_L is opposite to the defined polarity of Fig. 21.

The voltage then decays (with the same time constant as the current i_L) to zero because the inductor is approaching its short-circuit equivalence. The equation for v_L is therefore

$$v_L = V_m e^{-t/\tau} \implies v_L = -20 \text{ V} e^{-t/11.11 \mu\text{s}}$$

c. See Fig. 22 next slide. The initial and final values of the current were drawn first, and then the transient response was included between these levels. For the voltage, the waveform begins and ends at zero, with the peak value having a sign sensitive to the defined polarity of v_L in Fig. 21.

6. Initial Conditions (7/7)

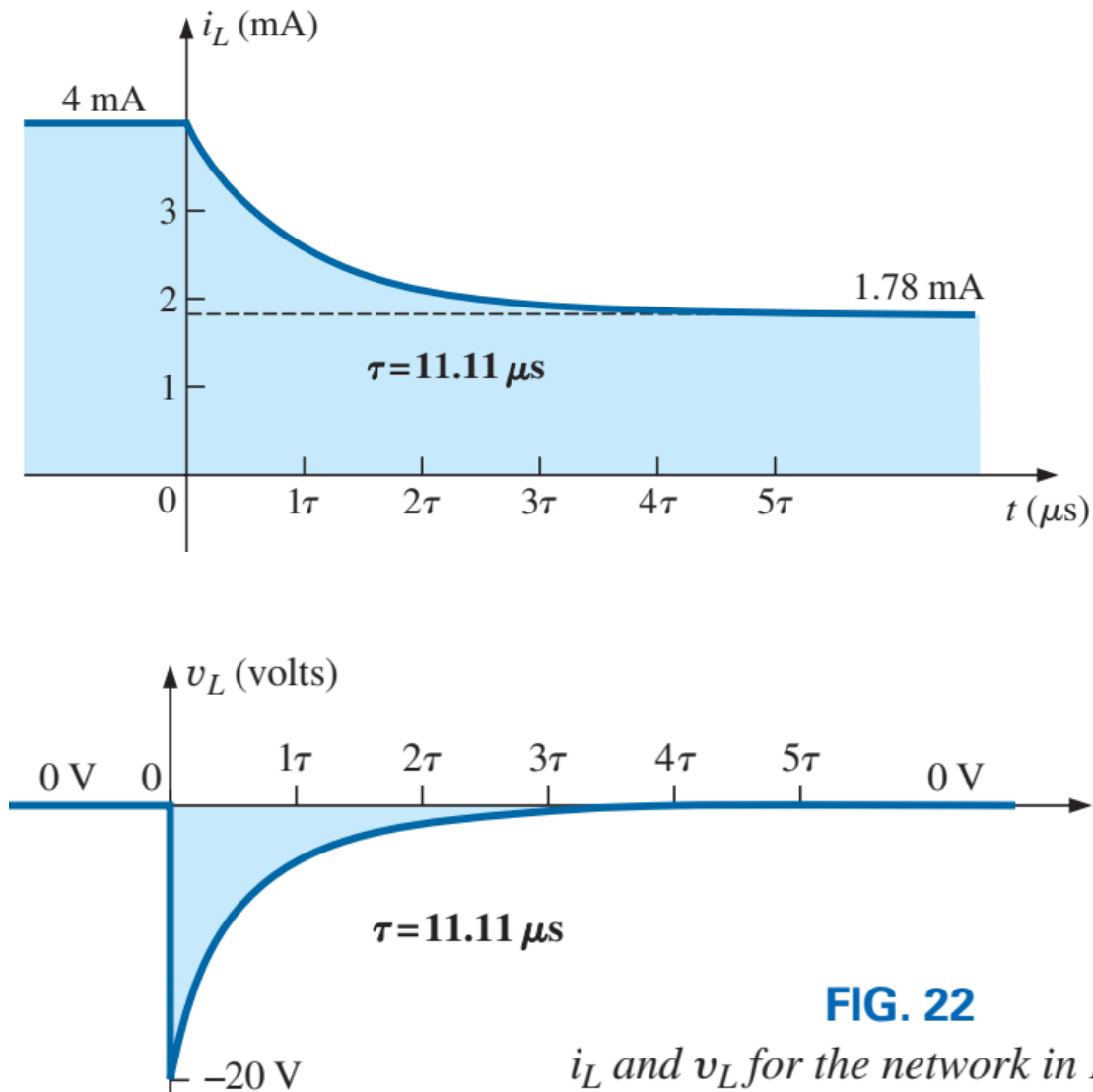


FIG. 22

i_L and v_L for the network in Fig. 21.

7. R - L Transients: The Release Phase (1/12)

- If the series R - L circuit in Fig.23 reaches steady-state conditions and the switch is quickly opened, a spark will occur across the contacts due to the rapid change in current from a maximum of E/R to zero amperes.

Apply KVL closewise in Fig. 23.

- $E + (-V_{\text{contact}}) + (-V_R) + (V_L) = 0$
- $V_R = i_R R = (0A)R = 0$
- $E - V_{\text{contact}} + (0V) + V_L = 0$
- $V_{\text{contact}} = E + V_L$

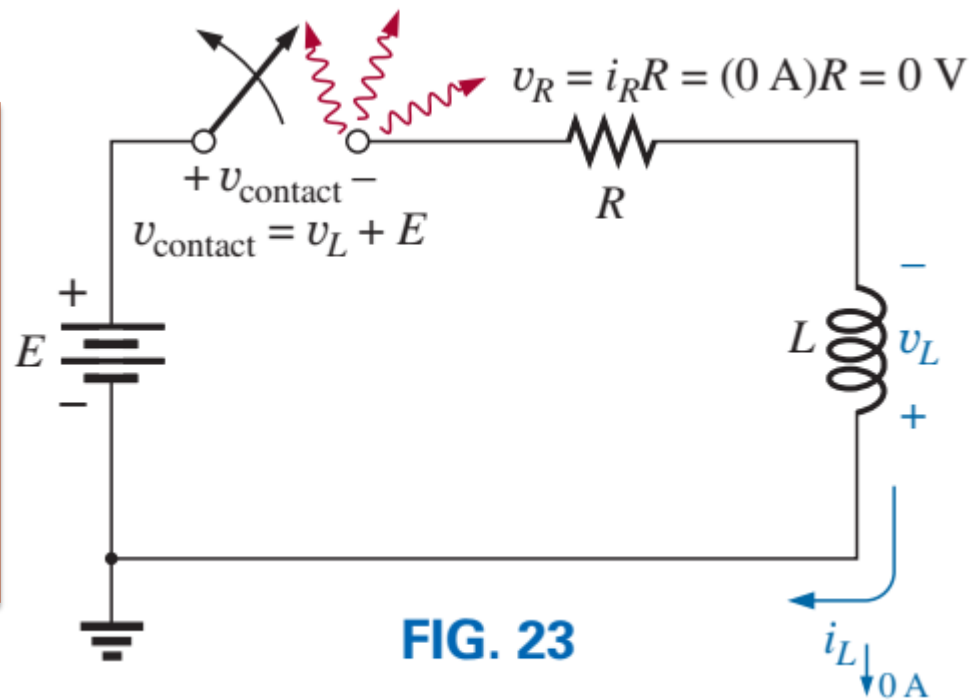


FIG. 23

7. R - L Transients: The Release Phase (2/12)

- The solution to an issue found in figure 23 is to use a network like that in Fig. 24(a). When the switch is closed, the voltage across resistor R_2 is E volts.
- A Thévenin network of E in parallel with R_2 results in the source as shown in Fig. 24(b), since R_2 will be shorted out by the short-circuit replacement of the voltage source E when the Thévenin resistance is determined. *This is explained in next slide.*

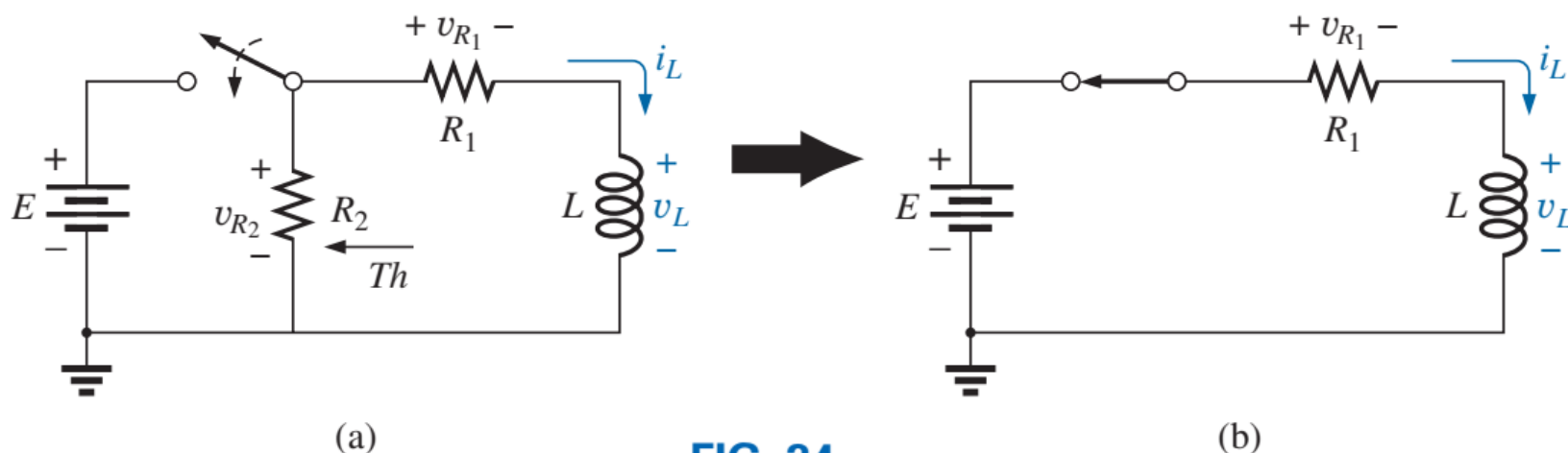


FIG. 24

7. R - L Transients: The Release Phase (3/12)

In Fig. 24(a), previous slide

- When switch is closed.
- $V_{R2} = E$ (V_{R2} and E are in parallel).

In Fig. 24(b), previous slide

- When switch is closed and in order to find R_{Th} , we make voltage source, $E = 0V$.
- As , $V_{R2} = E$, therefore $V_{R2} = 0V$.
- $V_{R2} = 0V$ ($R_2 = 0\Omega$, R_2 is shorted out).

7. R-L Transients: The Release Phase (4/12)

After the storage phase has passed and steady-state conditions are established, the switch can be opened without the sparking effect or rapid discharge due to resistor R_2 , which provides a complete path for the current i_L . In fact, for clarity the discharge path is isolated in Fig. 25. The voltage v_L across the inductor reverses polarity and has a magnitude determined by (*Apply KVL in clockwise direction in figure 25*)

$$-V_{R_2} - V_{R_1} - v_L = 0$$



$$v_L = -(v_{R_1} + v_{R_2}) \quad (18)$$

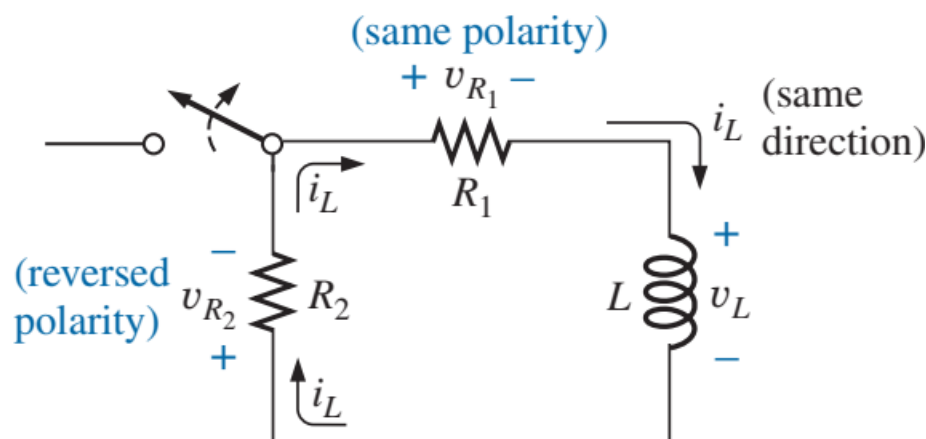


FIG. 25

7. R - L Transients: The Release Phase (5/12)

Recall that the voltage across an inductor can change instantaneously but the current cannot. The result is that the current i_L must maintain the same direction and magnitude, as shown in Fig. 25. Therefore, the instant after the switch is opened, i_L is still $I_m = E/R_1$, and

$$\begin{aligned} v_L &= -(v_{R_1} + v_{R_2}) = -(i_1 R_1 + i_2 R_2) \\ &= -i_L(R_1 + R_2) = -\frac{E}{R_1}(R_1 + R_2) = -\left(\frac{R_1}{R_1} + \frac{R_2}{R_1}\right)E \end{aligned}$$

- $I_m = i_L = i_1 = i_2$

and

$$v_L = -\left(1 + \frac{R_2}{R_1}\right)E$$

switch opened (19)

When the switch is opened, the voltage across the inductor reverses polarity and drops instantaneously from E to $-[1 + (R_2/R_1)]E$ volts.

7. R - L Transients: The Release Phase (6/12)

As an inductor releases its stored energy, the voltage across the coil decays to zero in the following manner:

$$v_L = -V_i e^{-t/\tau'} \quad (20)$$

with
$$V_i = \left(1 + \frac{R_2}{R_1}\right)E$$

and
$$\tau' = \frac{L}{R_T} = \frac{L}{R_1 + R_2}$$

The current decays from a maximum of $I_m = E/R_1$ to zero.

Using Eq. (17) gives

$$I_i = \frac{E}{R_1} \quad \text{and} \quad I_f = 0 \text{ A}$$

so that
$$i_L = I_f + (I_i - I_f)e^{-t/\tau'} = 0 \text{ A} + \left(\frac{E}{R_1} - 0 \text{ A}\right)e^{-t/\tau'}$$

and
$$i_L = \frac{E}{R_1} e^{-t/\tau'} \quad (21)$$

7. R - L Transients: The Release Phase (7/12)

with
$$\tau' = \frac{L}{R_1 + R_2}$$

The mathematical expression for the voltage across either resistor can then be determined using Ohm's law:

$$v_{R_1} = i_{R_1} R_1 = i_L R_1 = \frac{E}{R_1} R_1 e^{-t/\tau'}$$

and

$$v_{R_1} = E e^{-t/\tau'} \quad (22)$$

The voltage v_{R_1} has the same polarity as during the storage phase since the current i_L has the same direction. The voltage v_{R_2} is expressed as follows using the defined polarity of Fig. 24(a).

$$v_{R_2} = -i_{R_2} R_2 = -i_L R_2 = -\frac{E}{R_1} R_2 e^{-t/\tau'}$$

and

$$v_{R_2} = -\frac{R_2}{R_1} E e^{-t/\tau'} \quad (23)$$

7. R - L Transients: The Release Phase (8/12)

EXAMPLE 4 Resistor R_2 was added to the network in Fig. 18(slide 36) as shown in Fig. 26.

- Find the mathematical expressions for i_L , v_L , v_{R_1} , and v_{R_2} for five time constants of the storage phase.
- Find the mathematical expressions for i_L , v_L , v_{R_1} , and v_{R_2} if the switch is opened after five time constants of the storage phase.
- Sketch the waveforms for each voltage and current for both phases covered by this example. Use the defined polarities in Fig. 25.

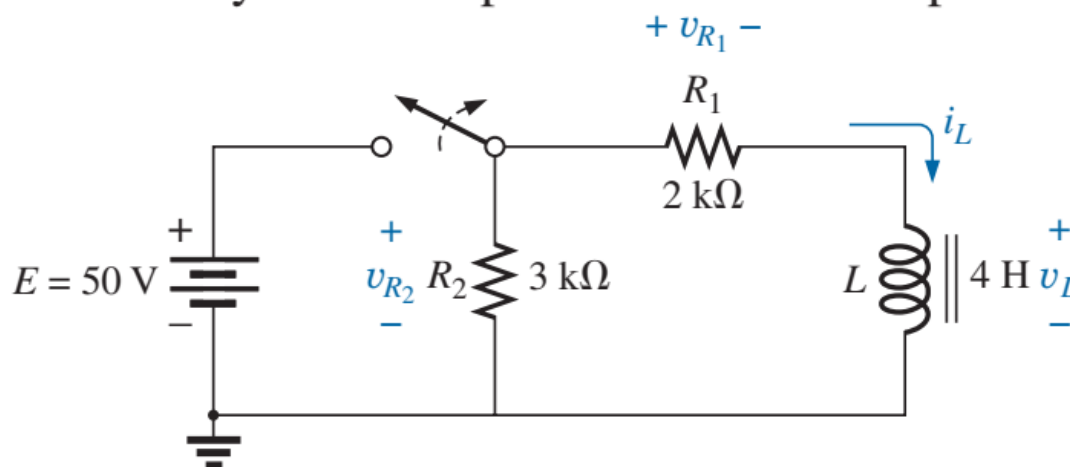


FIG. 26

7. *R-L* Transients: The Release Phase (9/12)

Solutions:

a. From Example 2:

$$i_L = 25 \text{ mA} (1 - e^{-t/2 \text{ ms}})$$

$$v_L = 50 \text{ V} e^{-t/2 \text{ ms}}$$

$$\begin{aligned} v_{R_1} &= i_{R_1} R_1 = i_L R_1 \\ &= \left[\frac{E}{R_1} (1 - e^{-t/\tau}) \right] R_1 \\ &= E(1 - e^{-t/\tau}) \end{aligned}$$

and

$$v_{R_1} = 50 \text{ V} (1 - e^{-t/2 \text{ ms}})$$

$$v_{R_2} = E = 50 \text{ V}$$

$$\begin{aligned} \text{b. } \tau' &= \frac{L}{R_1 + R_2} = \frac{4 \text{ H}}{2 \text{ k}\Omega + 3 \text{ k}\Omega} = \frac{4 \text{ H}}{5 \times 10^3 \Omega} \\ &= 0.8 \times 10^{-3} \text{ s} = 0.8 \text{ ms} \end{aligned}$$

By Eqs. (19) and (20):

$$V_i = \left(1 + \frac{R_2}{R_1} \right) E = \left(1 + \frac{3 \text{ k}\Omega}{2 \text{ k}\Omega} \right) (50 \text{ V}) = 125 \text{ V}$$

and

$$v_L = -V_i e^{-t/\tau'} = -125 \text{ V} e^{-t/-0.8 \text{ ms}}$$

7. *R-L* Transients: The Release Phase (10/12)

By Eq. (21):

$$I_m = \frac{E}{R_1} = \frac{50 \text{ V}}{2 \text{ k}\Omega} = 25 \text{ mA}$$

and
$$i_L = I_m e^{-t/\tau'} = \mathbf{25 \text{ mA} e^{-t/0.8 \text{ ms}}}$$

By Eq. (22):

$$v_{R_1} = E e^{-t/\tau'} = \mathbf{50 \text{ V} e^{-t/0.8 \text{ ms}}}$$

By Eq. (23):

$$v_{R_2} = -\frac{R_2}{R_1} E e^{-t/\tau'} = -\frac{3 \text{ k}\Omega}{2 \text{ k}\Omega} (50 \text{ V}) e^{-t/\tau'} = \mathbf{-75 \text{ V} e^{-t/0.8 \text{ ms}}}$$

7. *R-L* Transients: The Release Phase (11/12)

c. See Fig. 27.

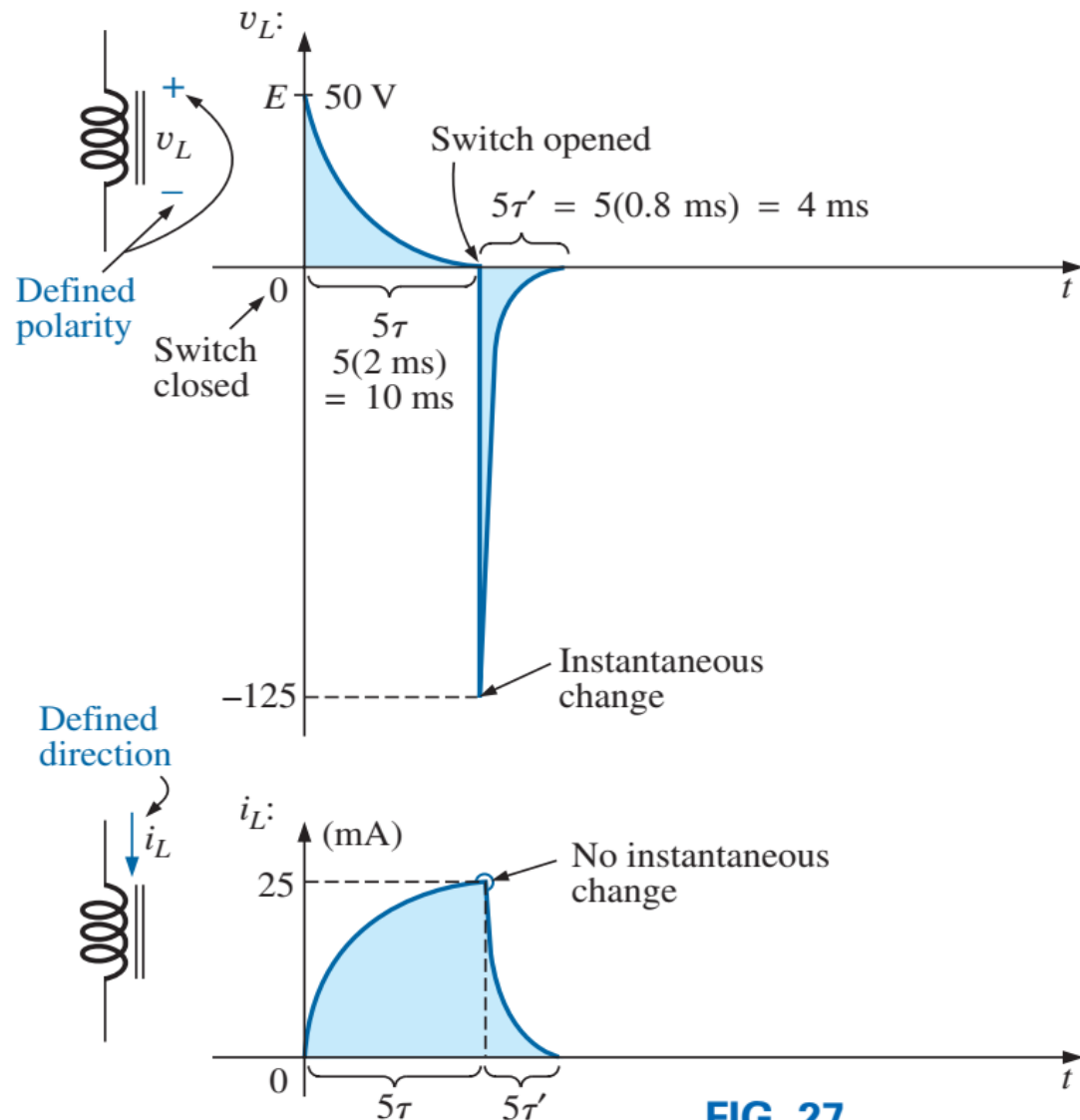


FIG. 27

7. *R-L* Transients: The Release Phase (12/12)

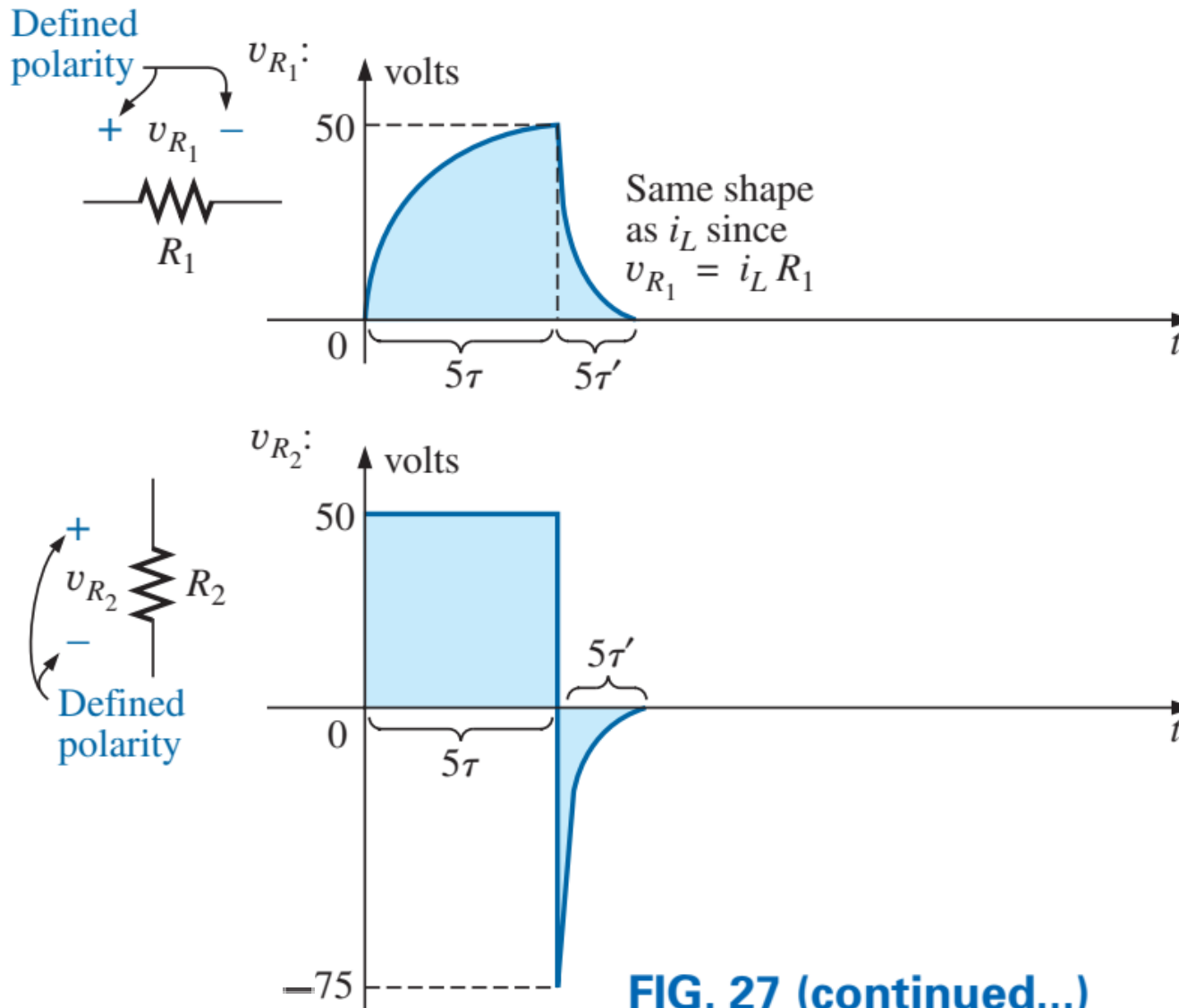


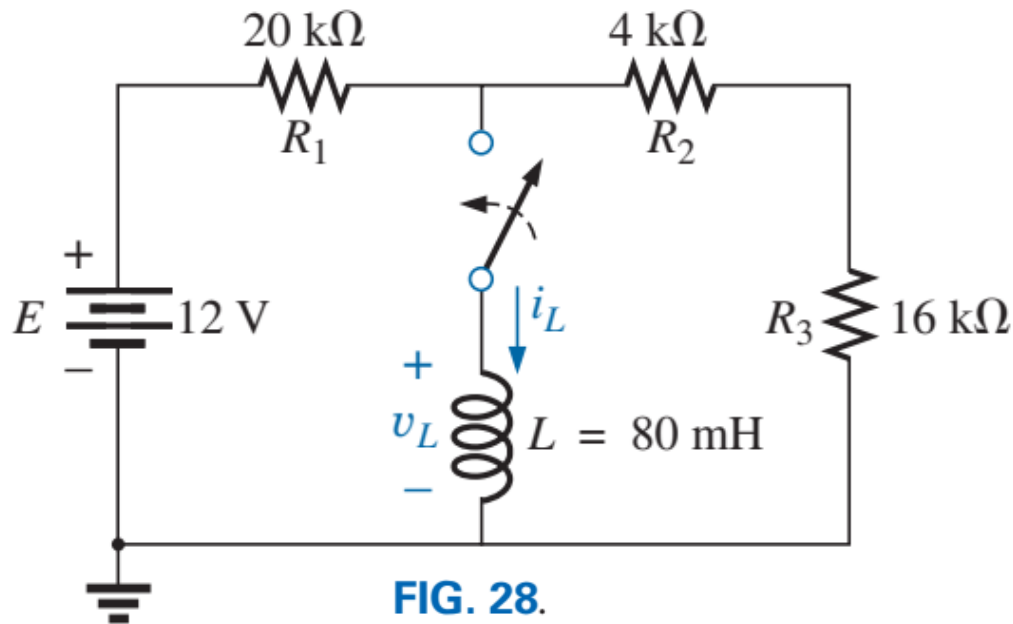
FIG. 27 (continued...)

8. Thevenin's Equivalent: $\tau = L/R_{Th}$ (1/4)

- A circuit does not always have the basic form in Fig. 13*

EXAMPLE 5 For the network in Fig. 28.

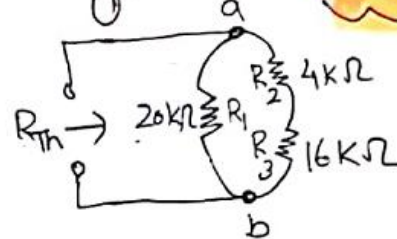
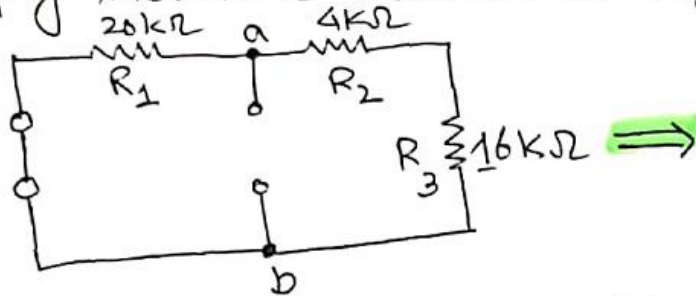
- Find the mathematical expression for the transient behavior of the current i_L and the voltage v_L after the closing of the switch ($I_i = 0$ mA).
- Draw the resultant waveform for each.



8. Thevenin's Equivalent: $\tau = L/R_{Th}$ (2/4)

Solution

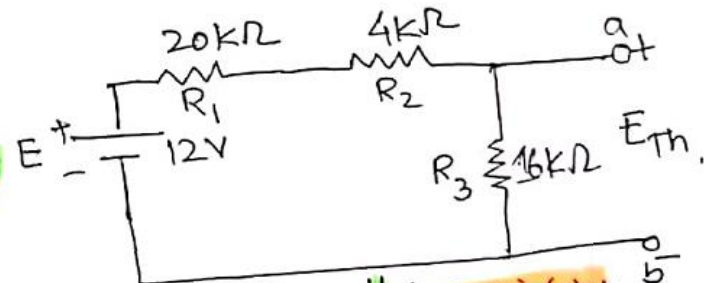
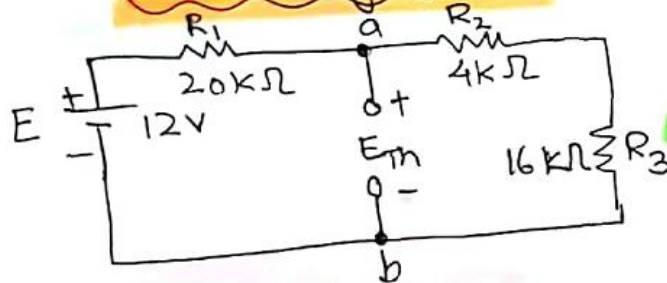
① Apply Thevenin's theorem to Fig 28 yields. Calculating R_{Th} as:



$$R_{Th} = R_1 \parallel (R_2 + R_3) = 20k\Omega \parallel (4k\Omega + 16k\Omega) = 20k\Omega \parallel 20k\Omega = 10k\Omega$$

$$R_{Th} = 10k\Omega$$

Calculating E_{Th} as:



↓ (As R_1 is in series with R_3 so $R' = R_2 + R_3$)

$$E_{Th} = \frac{R' E}{R_1 + R'} = \frac{(20k\Omega)(12)}{20k\Omega + 20k\Omega} = 6V$$

$$E_{Th} = 6V$$

8. Thevenin's Equivalent: $\tau = L/R_{Th}$ (3/4)

The Thévenin equivalent circuit is shown in Fig. 29. Using Eq. (13):

$$i_L = \frac{E_{Th}}{R} (1 - e^{-t/\tau})$$

$$\tau = \frac{L}{R_{Th}} = \frac{80 \times 10^{-3} \text{ H}}{10 \times 10^3 \Omega} = 8 \times 10^{-6} \text{ s} = 8 \mu\text{s}$$

$$I_m = \frac{E_{Th}}{R_{Th}} = \frac{6 \text{ V}}{10 \times 10^3 \Omega} = 0.6 \times 10^{-3} \text{ A} = 0.6 \text{ mA}$$

and $i_L = \mathbf{0.6 \text{ mA} (1 - e^{-t/8\mu\text{s}})}$

Using Eq. (15):

$$v_L = E_{Th} e^{-t/\tau}$$

so that

$$v_L = \mathbf{6 \text{ V} e^{-t/8\mu\text{s}}}$$

Thévenin equivalent circuit:

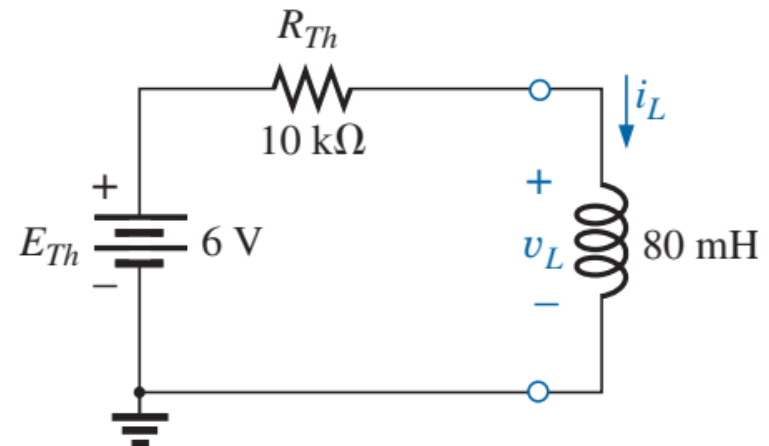


FIG. 29

8. Thevenin's Equivalent: $\tau = L/R_{Th}$ (4/4)

b. See Fig. 30.

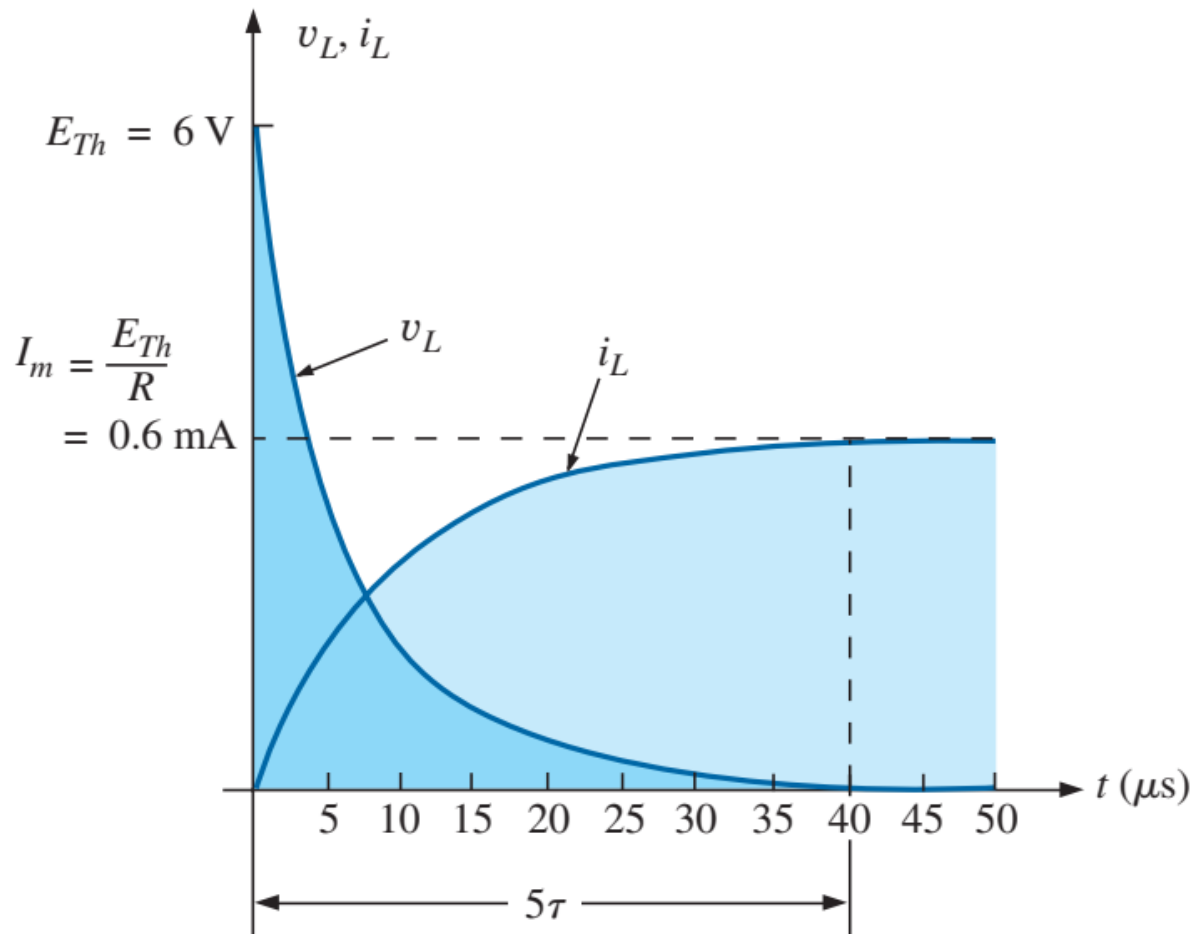


FIG. 30

9. Average Induced Voltage: $v_{L_{av}}$ (1/4)

- For inductors, the average induced voltage is defined by:

$$v_{L_{av}} = L \frac{\Delta i_L}{\Delta t} \quad (\text{volts, V}) \quad (24)$$

- Where Δ indicates a finite (measurable) change in current or time.

9. Average Induced Voltage: $v_{L_{av}}$ (2/4)

EXAMPLE 6 Find the waveform for the average voltage across the coil if the current through a 4 mH coil is as shown in Fig. 31.

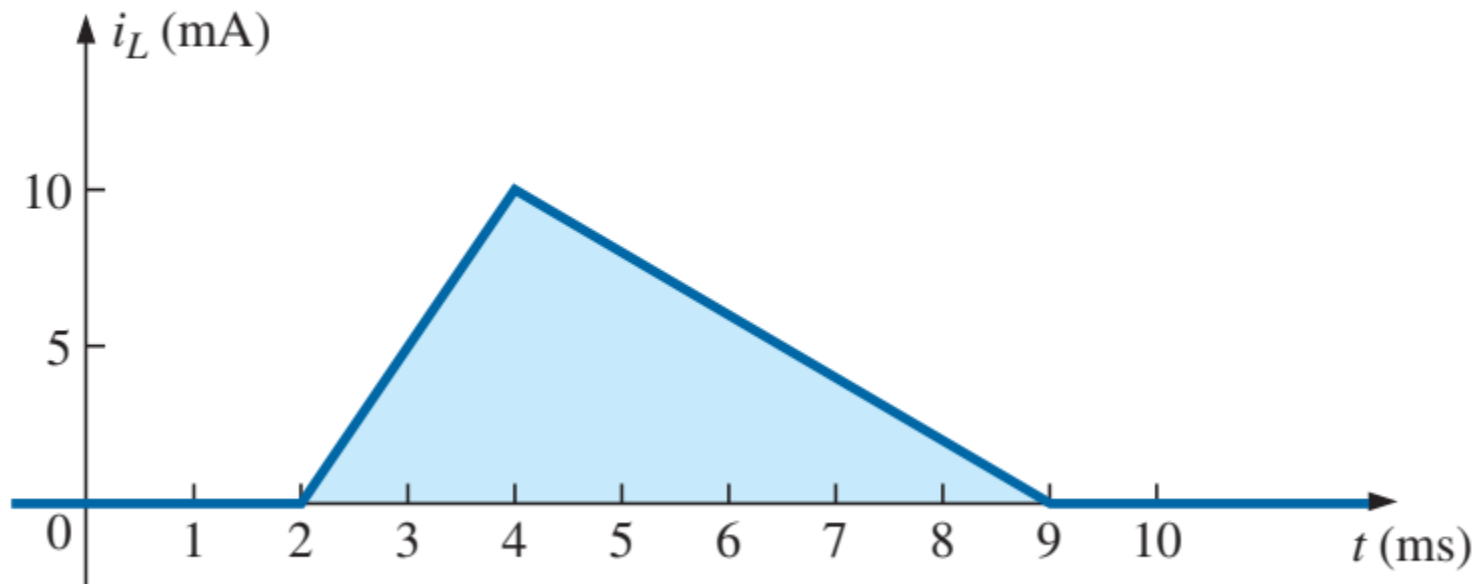


FIG. 31

9. Average Induced Voltage: $v_{L_{av}}$ (3/4)

Solutions:

- a. *0 to 2 ms*: Since there is no change in current through the coil, there is no voltage induced across the coil. That is,

$$v_L = L \frac{\Delta i}{\Delta t} = L \frac{0}{\Delta t} = \mathbf{0 \text{ V}}$$

- b. *2 ms to 4 ms*:

$$v_L = L \frac{\Delta i}{\Delta t} = (4 \times 10^{-3} \text{ H}) \left(\frac{10 \times 10^{-3} \text{ A}}{2 \times 10^{-3} \text{ s}} \right) = 20 \times 10^{-3} \text{ V} = \mathbf{20 \text{ mV}}$$

- c. *4 ms to 9 ms*:

$$v_L = L \frac{\Delta i}{\Delta t} = (-4 \times 10^{-3} \text{ H}) \left(\frac{10 \times 10^{-3} \text{ A}}{5 \times 10^{-3} \text{ s}} \right) = -8 \times 10^{-3} \text{ V} = \mathbf{-8 \text{ mV}}$$

- d. *9 ms to ∞* :

$$v_L = L \frac{\Delta i}{\Delta t} = L \frac{0}{\Delta t} = \mathbf{0 \text{ V}}$$

9. Average Induced Voltage: $v_{L_{av}}$ (4/4)

The waveform for the average voltage across the coil is shown in Fig. 32. Note from the curve that

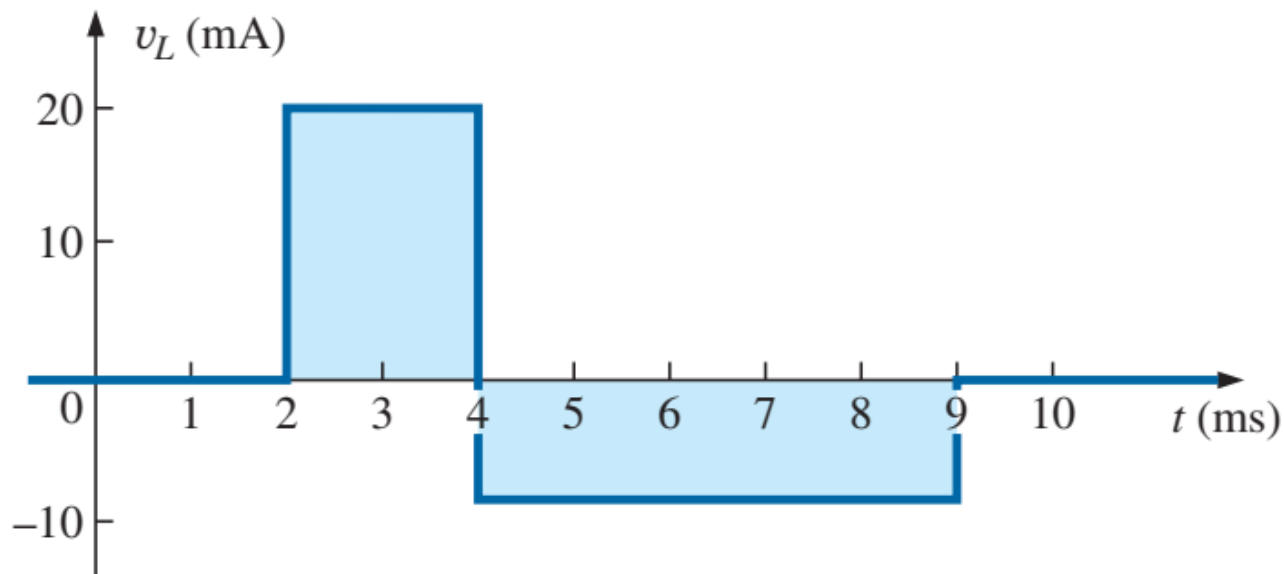


FIG. 32

10. Inductors in Series and in Parallel (1/2)

For inductors in series, the total inductance is found in the same manner as the total resistance of resistors in series (Fig. 33):

$$L_T = L_1 + L_2 + L_3 + \dots + L_N$$

25

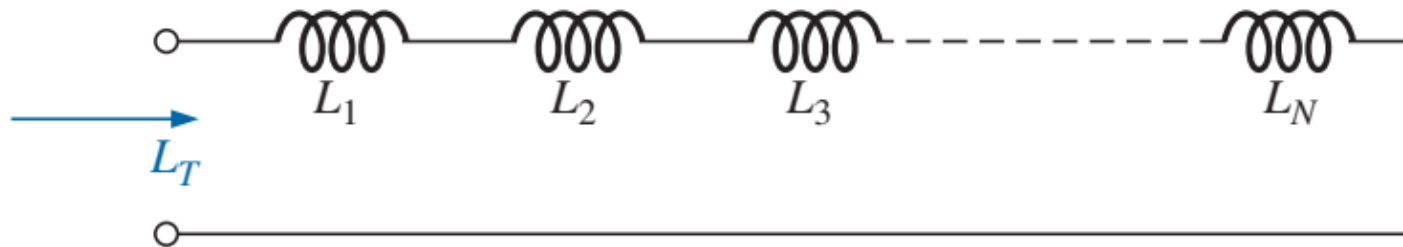


FIG. 33

Inductors in series.

10. Inductors in Series and in Parallel (2/2)

For inductors in parallel, the total inductance is found in the same manner as the total resistance of resistors in parallel (Fig. 34):

$$\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \cdots + \frac{1}{L_N} \quad 26$$

For two inductors in parallel,

$$L_T = \frac{L_1 L_2}{L_1 + L_2} \quad 27$$

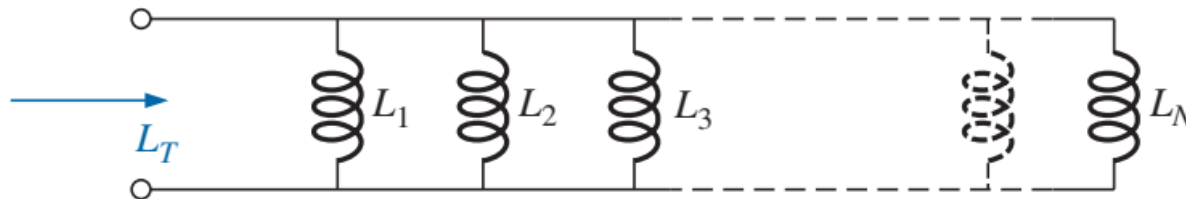
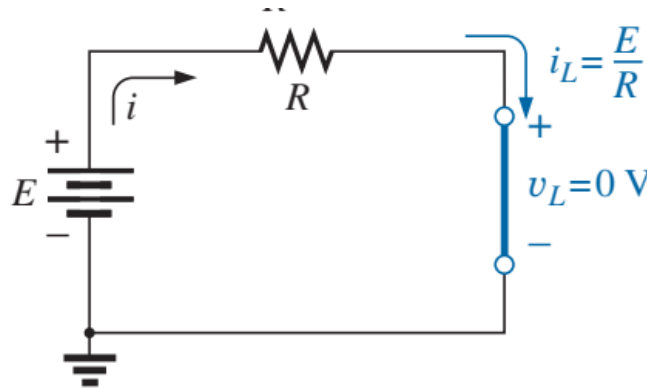


FIG. 34

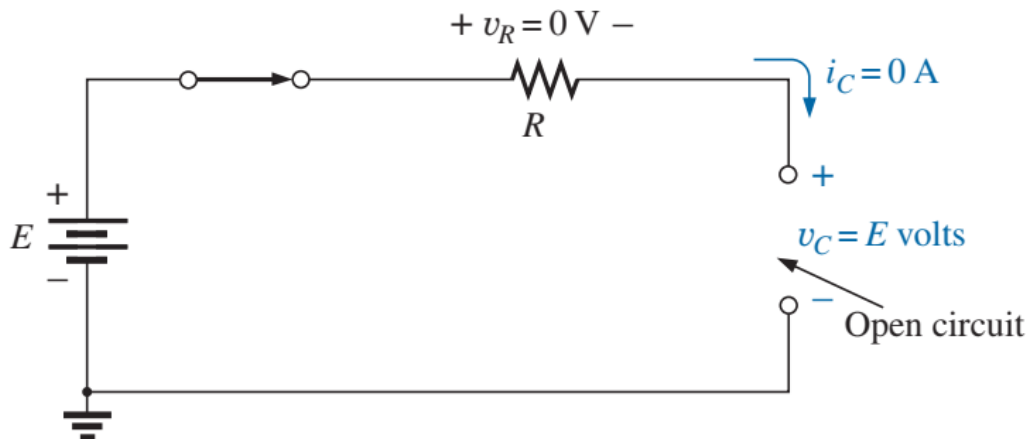
Inductors in parallel.

11. Steady-State Conditions (1/3)

- The inductor takes on the characteristics of a short circuit when steady-state conditions have been established.*



The capacitor takes on the characteristics of an open circuit when steady-state conditions have been established.



11. Steady-State Conditions (2/3)

EXAMPLE 7 Find the current I_L and the voltage V_C for the network in Fig. 35.

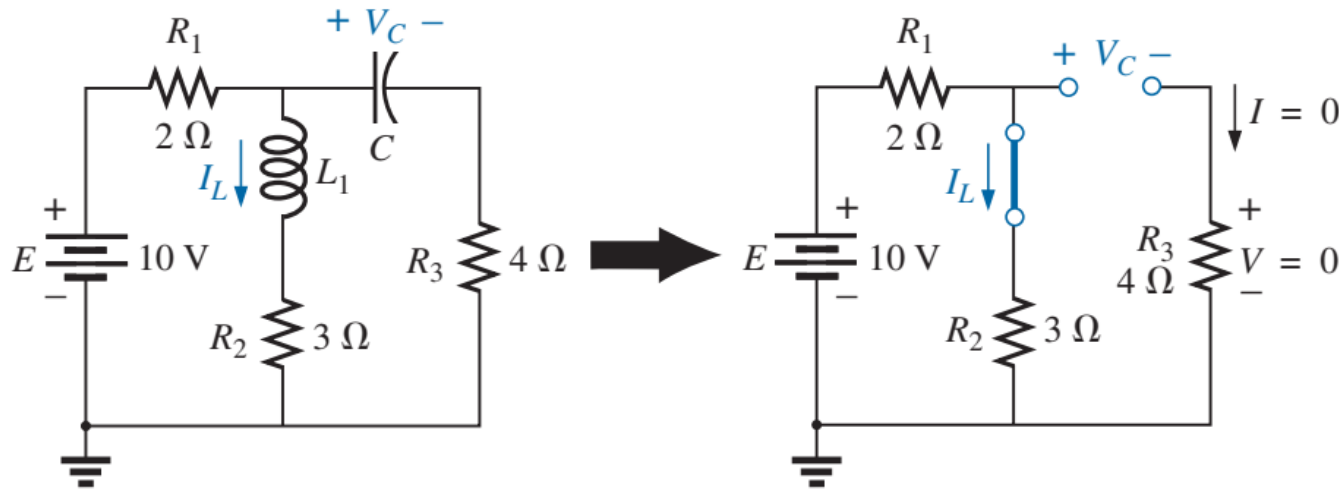


FIG. 35

Solution:

$$I_L = \frac{E}{R_1 + R_2} = \frac{10 \text{ V}}{5 \Omega} = 2 \text{ A}$$

$$V_C = \frac{R_2 E}{R_2 + R_1} = \frac{(3 \Omega)(10 \text{ V})}{3 \Omega + 2 \Omega} = 6 \text{ V}$$

11. Steady-State Conditions (3/3)

Home-Assignment

EXAMPLE 8 Find currents I_1 and I_2 and voltages V_1 and V_2 for the network in Fig. 36.

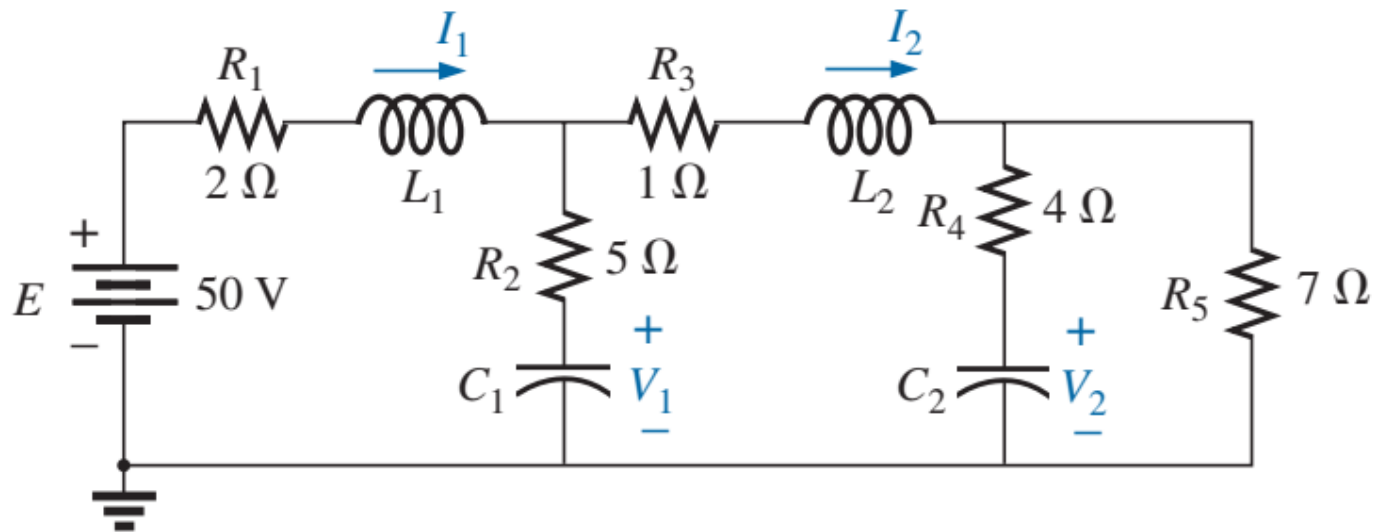


FIG. 36.

12. Energy stored by an Inductor (1/3)

- The ideal inductor, like the ideal capacitor, does not dissipate the electrical energy supplied to it. It stores the energy in the form of a magnetic field. A plot of the voltage, current, and power to an inductor is shown in Fig. 37 during the build-up of the magnetic field surrounding the inductor. The energy stored is represented by the shaded area under the power curve. Using calculus, we can show that the evaluation of the area under the curve yields:

$$W_{\text{stored}} = \frac{1}{2}LI_m^2 \quad (\text{joules, j}) \quad (28)$$

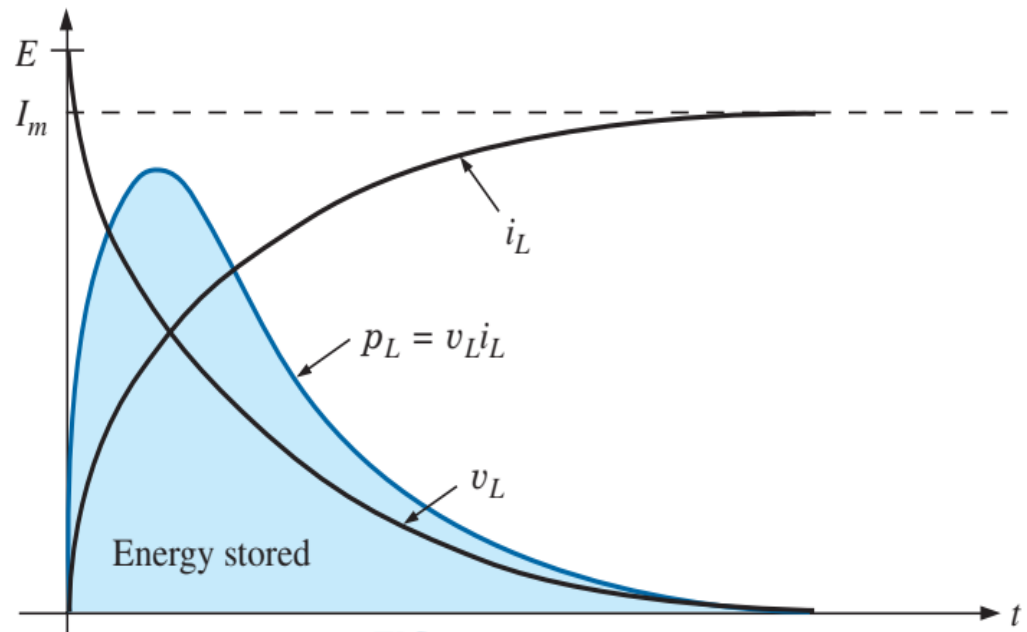


FIG. 37

12. Energy stored by an Inductor (2/3)

EXAMPLE 9 Find the energy stored by the inductor in the circuit in Fig. 38 when the current through it has reached its final value.

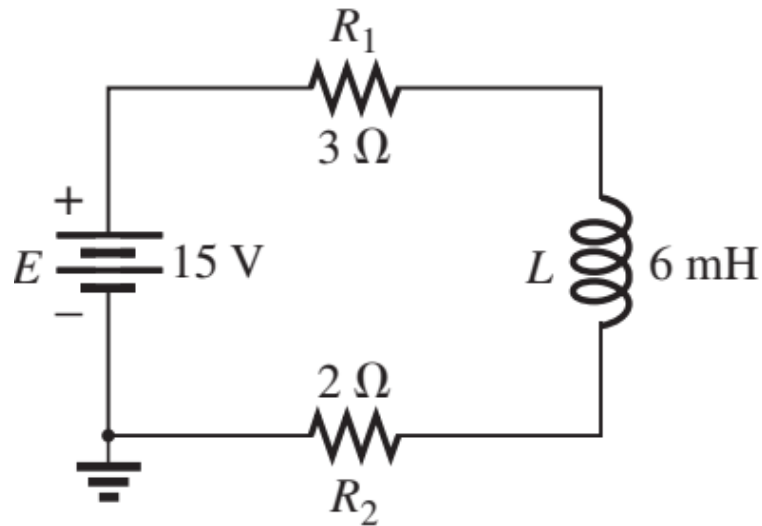
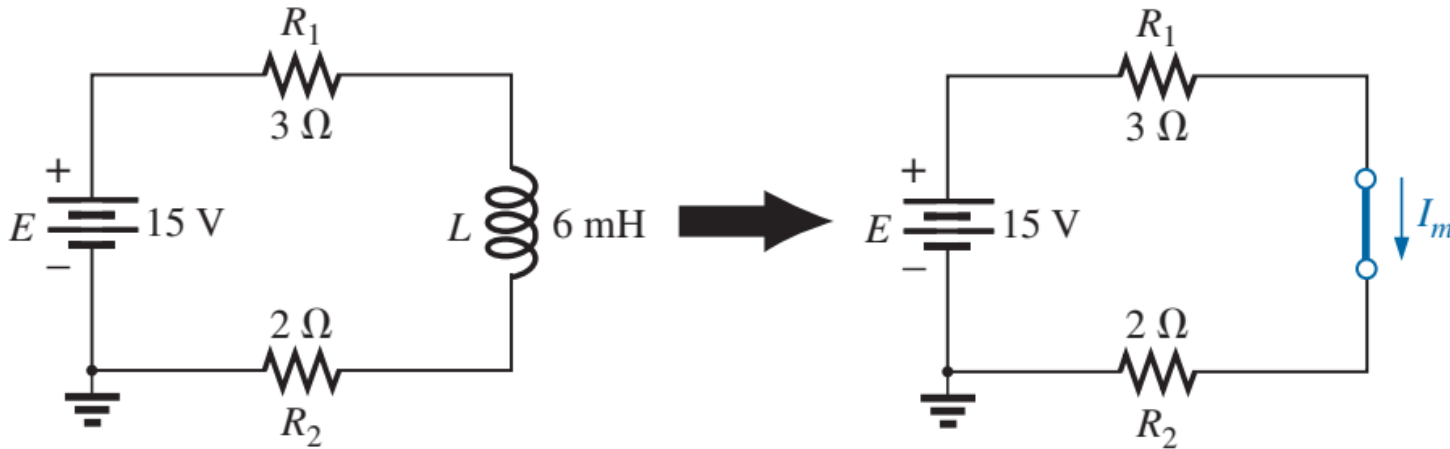


FIG. 38

12. Energy stored by an Inductor (3/3)

Solution:



$$I_m = \frac{E}{R_1 + R_2} = \frac{15\text{ V}}{3\ \Omega + 2\ \Omega} = \frac{15\text{ V}}{5\ \Omega} = 3\text{ A}$$

$$W_{\text{stored}} = \frac{1}{2}LI_m^2 = \frac{1}{2}(6 \times 10^{-3}\text{ H})(3\text{ A})^2 = \frac{54}{2} \times 10^{-3}\text{ J} = \mathbf{27\text{ mJ}}$$

References: Textbooks

1. *Robert L.Boylestad. (2014). Introductory Circuit Analysis. 12th Edition. Pearson Education Limited.*
2. *Allan H.Robbins, Wilhelm C.Miller. (2012). Circuit Analysis Theory and Practice. 5th Edition. Cengage Learning.*
3. *Charles K.Alexander, Matthew N.O.Sadiku (2017). Fundamentals of Electric Circuits. 6th Edition. McGraw Hill Education.*